

FactorJEPA: Factorized Predictor Surgery for World Models in Dense, Heterogeneous Urban Scenes

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Abstract

World models aim to capture and predict the **structure and dynamics of the physical world**, and **Joint Embedding Predictive Architectures (JEPA)** are a compelling direction. We study a largely unexplored regime: **populous, crowded, and chaotic Global South urban environments**, which we call **DENSEWORLD**, where **soft spatial boundaries, extreme agent heterogeneity, persistent occlusion, and rapid social negotiation** replace the lane-structured assumptions behind existing evaluations. We release **DENSEWORLD 1.0: 115,687 videos (276 hours)** of drive-through, walk-through, and aerial footage, each 4 to 10 seconds, spanning **22 Indian cities** and **714 long-form sources**, of which **86,831** carry automatically derived *layout*, *agent*, and *interaction* targets. **Ten off-the-shelf frozen encoders** (DINOv2, I-JEPA, LeJEPA, and V-JEPA 1/2.0/2.1) all collapse into a narrow **37.5% to 44.4%** band on an 11-way motion-class probe with a 19.5% majority baseline, whereas DENSEWORLD-adapted encoders reach **50.3% to 53.2%**.

We introduce FactorJEPA, which makes *world structure a first-class predictive primitive*. Rather than adapting a JEPA predictor on undifferentiated video, it performs *predictor surgery*: a three-stage curriculum over automatically separated *layout*, *agent*, and *interaction* views, updating at most 8/48 (2B) and 6/40 (1B) encoder blocks with the predictor. Trained on **86,831** factor-annotated videos and evaluated on **23,106** held-out samples (1B V-JEPA 2.1), it cuts *future-latent L1* by **10.6%**, *causal future-block L1* by **8.9%** and *mask-ratio slope* by **27.4%** against the frozen backbone, and raises *motion-cosine separation* to **15.3×** the frozen value, each gain exceeding **70×** the paired 95% confidence interval. On both prediction diagnostics it separates in its favour against **all six** adaptation techniques we evaluate, namely full fine-tuning, linear probing then fine-tuning (LP-FT), low-rank adaptation (LoRA), weight-decomposed low-rank adaptation (DoRA), automatic relative-gradient-norm block selection (Auto-RGN), and vanilla continual self-supervised learning (continual-SSL), at **both** backbone scales, giving **24 of 24** paired wins, and beats the strongest full-scale baseline, LoRA, by **+2.8%**, **+1.6%**, **+25.2%** and **+17.8%**.

20-arm (2B) and 14-arm (1B) studies show rankings **replicate across backbone scale** at $\rho = 0.895$ to 0.978 , establishing the 1B model as a **half-cost proxy** for 2B selection. On a

10k screening subset, factorization concedes maximally linear motion readout to full fine-tuning (-44.9% at 2B, -33.4% at 1B), a trade-off that disappears at full scale, where FactorJEPA leads motion by $+25.2\%$ and gains more than LoRA on **all four** diagnostics. We release the dataset construction pipeline, the adapted checkpoints, and the complete 15-diagnostic evaluation harness.

DENSEWORLD: A Benchmark for Populous, Crowded, and Chaotic Global South

We use **DENSEWORLD** to denote a **critical but under-represented world-modeling regime: populous, crowded, and chaotic Global South urban scenes**. It is defined not only by geography, but by measurable properties: **high agent count density, high agent occupancy, persistent occlusion, agent heterogeneity, and interaction pressure**. This regime matters because many emerging urban environments combine **high population density, mixed mobility systems, diverse infrastructure patterns, and rapid spatial change**, making them scientifically important yet underrepresented in world-model benchmarks (Mahendra et al. 2021; Ellis and Roberts 2016). Thus, world models must handle uncertainty, density, and relational complexity that is less prominent in lower-density, lane-structured benchmarks. See Figure 2 for representative DENSEWORLD scenes.

We characterize DENSEWORLD through four coupled properties. **First**, it exhibits **soft spatial boundaries**: road edges, drivable space, and pedestrian regions are often negotiated rather than crisply marked (Varma et al. 2019; Dokania et al. 2023). **Second**, it contains **extreme agent heterogeneity**, with pedestrians, cars, two-wheelers, carts, animals, delivery vehicles, and rickshaws coexist locally (Khan and Maini 1999; Asaithambi et al. 2012). **Third**, it has **persistent occlusion** from density, clutter, and near-field encounters, producing severe **partial observability** (Varma et al. 2019; Dokania et al. 2023); prediction must rely on *memory*, *object permanence*, and *relational inference*, not only visible appearance. **Fourth**, motion is governed less by rigid rule compliance than by rapid **social negotiation** under mixed traffic and weak lane discipline (Papathanasopoulou and Antoniou 2018; Asaithambi et al. 2012).

Together, these properties make DENSEWORLD a **natural stress test** for world models: success requires repre-

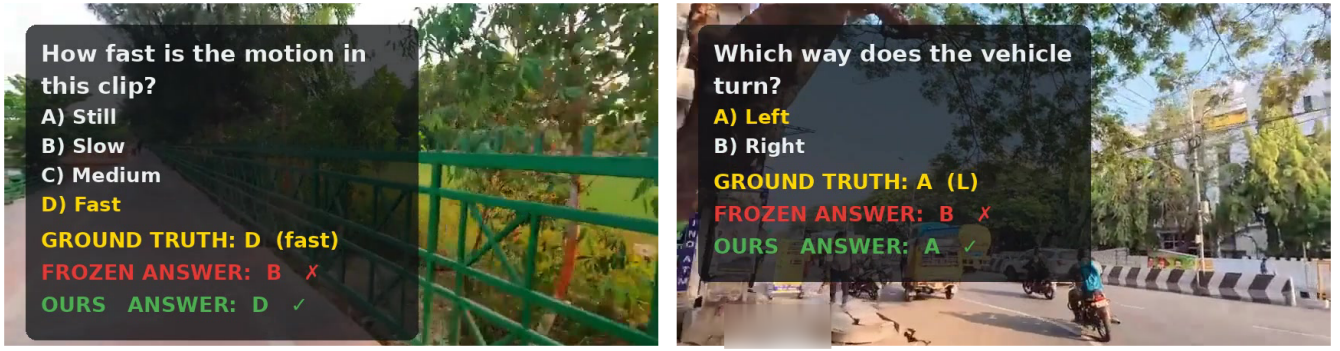


Figure 1: **Same question, two encoders.** Each held-out clip is posed as a multiple-choice question with a ground-truth answer, then answered twice from an identical attentive probe head: once over the **frozen** V-JEPA 2.1 encoder and once over **ours** (factor-view predictor surgery). Only the backbone differs. *Left*: motion-speed quartile from optical flow (ours 69.8% vs frozen 60.9% over 1,825 held-out clips; chance 25.6%). *Right*: vehicle turn direction under a leave-one-video-out split. Selected qualitative examples.

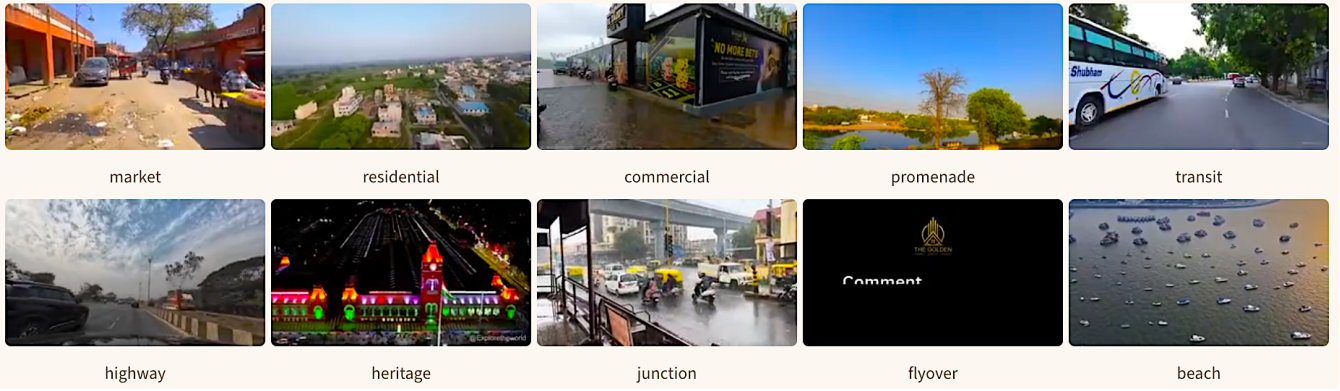


Figure 2: **DENSEWORLD 1.0 scene-type coverage.** Representative examples from the dataset illustrate the breadth of urban environments covered by DENSEWORLD 1.0, including **market**, **residential**, **commercial**, **promenade**, **transit**, **highway**, **heritage**, **junction**, **flyover**, **beach**, **ghat**, **bazaar**, and **skyline** scenes. This diversity reflects the spatial, social, and infrastructural heterogeneity of populous, crowded, and chaotic Global South urban environments.

sentations that preserve **layout**, **agent state**, and **interaction dynamics** under uncertainty, rather than collapsing into one **entangled predictive latent**. As shown in Section , we measure this regime through agent count density, agent occupancy, occlusion pressure, interaction pressure, and agent heterogeneity. This motivates a predictor that separates **lay-out**, **entities**, and **interactions**.

DENSEWORLD 1.0: the Benchmark

We collected **714** long-form YouTube videos at 480p, totalling **276 hours** of urban footage across **22 Tier-1** and **Tier-2 cities in India**. The data spans **drive-through**, **walk-through**, and **aerial** viewpoints, and long-form recordings are segmented using scene-detection and shot-boundary methods into short videos of **4 to 10 seconds**. DENSEWORLD 1.0 covers commercial, residential, transit, heritage, coastal, and high-density street settings, including **markets**, **commercial streets**, **transit corridors**, **junctions**, **flyovers**, **bazaars**, **ghats**, **beaches**, and **skylines**. It further spans variations in the **time of day**, **weather**, **crowd density**, **traffic mix**, **pedestrian-vehicle separation**, **road geometry**, **sur-**

face quality, **encroachment**, and **lighting**. Beyond standard urban actors, the dataset includes locally frequent agents and objects such as **auto-rickshaws**, **cycle rickshaws**, **street vendors**, and **animals**. Before release, all footage is processed through privacy-preserving filters, including face and license-plate blurring and removal of sensitive segments.

A defining feature of DENSEWORLD is **heterogeneous mixed traffic**. In many developing-country cities, multiple vehicle and actor types share the same right of way under **weak lane discipline**, with strong **lateral** and **longitudinal** coupling (Khan and Maini 1999; Papathanasopoulou and Antoniou 2018; Asaithambi et al. 2012). This is not a mild variant of standard traffic, but a **qualitatively different operating regime**. Prior work highlights the coexistence of **cars**, **buses**, **trucks**, **auto-rickshaws**, **scooters**, **motorcycles**, **bicycles**, **carts**, and other slow- and fast-moving agents within the same corridor (Khan and Maini 1999; Asaithambi et al. 2012). Here, scene structure does not decompose into *static lanes*, *isolated actors*, and *simple pairwise relations*; spatial support is **fluid**, visibility is unstable, and behavior is shaped by dense **local negotiation**.

How Dense is DENSEWORLD? Comparison with Standard Driving Benchmarks

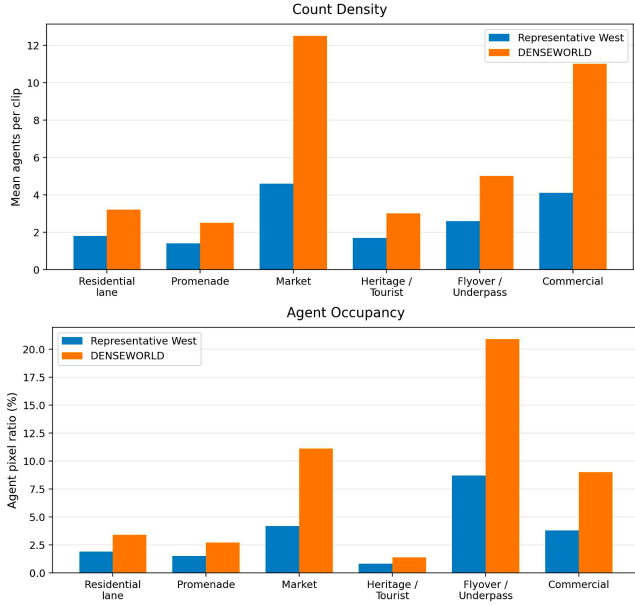


Figure 3: **DENSEWORLD exhibits higher multi-agent density than standard driving benchmarks.** We compare **agent count density** and **agent occupancy** across matched scene types. DENSEWORLD shows large gaps in **market**, **commercial**, and **flyover/underpass** scenes, indicating stronger interaction pressure and heavier visual competition.

To make this regime measurable, we compare DENSEWORLD against **BDD100K** and **nuScenes**, two widely used benchmarks for autonomous-driving perception and scene understanding (Yu et al. 2020; Caesar et al. 2020). The goal is not to claim that DENSEWORLD scenes merely look different, but to test whether they occupy a quantitatively distinct operating regime for world-model evaluation.

We measure this regime along five axes: **agent count density**, **agent occupancy**, **occlusion pressure**, **interaction pressure**, and **agent heterogeneity**. These metrics capture, respectively, the number of dynamic agents, the fraction of image area they occupy, the frequency of partial or heavy occlusion, the density of spatially proximate agent pairs, and the diversity of co-occurring actor categories. Together, they quantify the multi-agent load, visual congestion, visibility degradation, local interaction structure, and traffic diversity of each scene.

For an auditable comparison, all statistics are computed after mapping labels to a **shared dynamic-agent taxonomy** and matching comparable scene categories across datasets. Figure 3 shows that DENSEWORLD has consistently higher density and occupancy across matched categories, with the largest gaps in **market**, **commercial**, and **flyover/underpass** scenes. These results support the central premise of the benchmark: DENSEWORLD is not merely a geographic extension of existing driving data; it defines a higher-density, higher-occlusion, and higher-interaction regime for evaluating predictive world models. Full automatic-annotation de-

tails, including thresholds, annotation taxonomy, export format, and loss mapping, are provided in Appendix C.

Can Fine-Tuning Close the DENSEWORLD Gap?

Before introducing FactorJEPA, we ask whether conventional adaptation can recover the predictive structure required by DENSEWORLD without reorganizing the JEPA predictor. We evaluate three complementary strategies: **LoRA** (Hu et al. 2022), **DoRA** (Liu et al. 2024b), and **Auto-RGN** (Lee et al. 2023). All use the same raw clips, optimization steps, optimizer, masking policy, and evaluation protocol; only the parameter-update mechanism changes.

Let x denote a video clip; M_c and M_t the context and target masks; f_θ the online encoder; $\bar{f}_\theta$ the momentum target encoder; and g_ϕ the predictor. All baselines retain the executed V-JEPA objective:

$$\mathcal{L}_{\text{JEPA}} = \mathbb{E}_x \left[\left\| g_\phi(f_\theta(M_c \odot x), M_t) - \text{sg}(\bar{f}_\theta(M_t \odot x)) \right\|_2^2 \right],$$

where sg denotes stop-gradient. This data- and step-matched protocol isolates adaptation capacity without introducing a different prediction target or training signal.

Table 1: **Protocol-matched V-JEPA adaptation baselines.** All methods retain the JEPA objective and train on the same raw clips; only the adaptation mechanism changes.

Method	Adaptation mechanism	Question tested
LoRA (Hu et al. 2022)	Adds low-rank updates, $\Delta W = (\alpha/r)BA$, to selected frozen projections.	Is parameter-efficient low-rank adaptation sufficient?
DoRA (Liu et al. 2024b)	Separates weight magnitude from direction and applies low-rank updates to the directional component.	Does weight decomposition recover structure missed by LoRA?
Auto-RGN (Lee et al. 2023)	Selects transformer blocks using their relative gradient norms and updates only the selected subset.	Can gradient-guided selection localize the required adaptation?

For LoRA, an adapted weight matrix is

$$W' = W_0 + \frac{\alpha}{r}BA, \quad B \in \mathbb{R}^{d_{\text{out}} \times r}, \quad A \in \mathbb{R}^{r \times d_{\text{in}}},$$

where $r \ll \min(d_{\text{in}}, d_{\text{out}})$. DoRA further separates the magnitude and direction of each weight vector:

$$W' = m \odot \frac{W_0 + \Delta W}{\|W_0 + \Delta W\|}, \quad \Delta W = \frac{\alpha}{r}BA,$$

allowing directional adaptation without coupling it to weight magnitude.

Following the relative-gradient-norm criterion of Lee et al. (2023), our blockwise Auto-RGN implementation scores transformer block ℓ as

$$s_\ell = \frac{\|\nabla_{\theta_\ell} \mathcal{L}_{\text{JEPA}}\|_2}{\|\theta_\ell\|_2 + \epsilon}, \quad \mathcal{S}_K = \text{TopK}_\ell(s_\ell),$$

and updates only $\{\theta_\ell : \ell \in \mathcal{S}_K\}$. Normalization by parameter magnitude prevents larger blocks from being favored

solely because of scale. LoRA and DoRA target the same projection families, while Auto-RGN receives a matched trainable-parameter budget. Ranks, scaling coefficients, selected projections, block budgets, trainable parameter counts, and measured GPU-hours are reported in Appendix C.

FactorJEPA: Predictor Surgery over Factorized Views

Motivation. Let x be a video clip, M_c and M_t its context and target masks, f_θ the online encoder, and $\bar{f}_\theta$ the momentum target encoder. A conventional JEPA predictor g_ϕ learns

$$g_\phi : (f_\theta(M_c \odot x), M_t) \mapsto \hat{Y}_{t+\Delta}, \quad Y_{t+\Delta}^* = \text{sg}[\Pi_{M_t} \bar{f}_\theta(x)],$$

by matching $\hat{Y}_{t+\Delta}$ to $Y_{t+\Delta}^*$. The objective specifies *what* to predict, but every clip presents layout, agents and interactions simultaneously, so nothing constrains *which* of them the predictor actually uses.

This ambiguity is consequential in **DENSEWORLD**, where layout, agent density, visibility and interaction pressure are strongly correlated. A high-capacity predictor can exploit *shortcut mixtures* — crowd texture as a proxy for interaction, visible appearance for object persistence, or road geometry for motion — without recovering the structure governing scene evolution.

FactorJEPA addresses this on the *input* side rather than by redesigning the predictor. We construct three complementary views of every clip, in which layout, agents and interactions are isolated by construction, and adapt a pretrained V-JEPA predictor over them as a staged curriculum. The architecture is unchanged; what changes is the sequence of distributions the predictor is asked to model, and how much of the encoder is free to move at each stage.

Automatic Factor-View Construction

For each privacy-filtered clip x_n we run an open-vocabulary detector (Grounding DINO, Swin-B) (Liu et al. 2024a) to localise dynamic agents, and propagate those detections through the clip with a promptable video segmenter (SAM 3) (Ravi et al. 2024). Detections are re-seeded at four anchor frames and propagated within each anchor’s segment, which keeps identities stable under the heavy occlusion typical of the regime. This yields, per clip, a set of agent masks and temporally associated tracklets

$$\mathcal{A}(x_n) = \left\{ \tau_n^{(i)} = (m_{n,t}^{(i)}, b_{n,t}^{(i)})_{t=t_s^{(i)}}^{t_e^{(i)}} \right\}_i.$$

A deterministic map converts these into three *views* of the same clip, each a video the predictor can be trained on directly:

- D_L (**layout**): agent regions are suppressed, leaving slowly varying spatial support — road geometry, buildings, drivable area.
- D_A (**agents**): background context is suppressed, leaving the dynamic actors and their motion.
- D_I (**interactions**): spatio-temporal tubes around co-occurring, spatially proximate tracklet pairs, isolating local multi-agent coupling.

Views are derived only from the training split; no detector or segmenter output is used as an evaluation label. Checkpoints, prompt vocabulary, association rules, thresholds and per-view coverage are reported in Appendix C.

Three-Stage Curriculum

Training proceeds over the three views in order of increasing relational complexity, $D_L \rightarrow D_A \rightarrow D_I$, with a mixture weight over views at each stage. The ordering is deliberate: layout is the most stationary signal and anchors the encoder before agent appearance and finally interaction structure are introduced.

Encoder capacity is released progressively. At stage s only the deepest K_s transformer blocks receive gradients,

$$\Theta_{\text{train}}^{(s)} = \Theta_{\text{pred}} \cup \Theta_{\text{top-}K_s}, \quad K_s = \lfloor \kappa_s L \rfloor,$$

for an encoder of depth L , so that at most 8/48 blocks (ViT-G) and 6/40 blocks (ViT-g) ever move. The lower blocks and the momentum target network follow the frozen / momentum protocol of the underlying V-JEPA implementation.

Training Objective

FactorJEPA retains the executed V-JEPA objective and adds three auxiliary terms:

$$\mathcal{L} = \mathcal{L}_{\text{JEPA}} + \beta \mathcal{L}_{\text{probe}} + \gamma \mathcal{L}_{\text{motion}} + \lambda \mathcal{L}_{\text{drift}}$$

where $\mathcal{L}_{\text{JEPA}}$ is the masked future-latent loss above, $\mathcal{L}_{\text{probe}}$ is a lightweight multi-task read-off used for checkpoint selection, $\mathcal{L}_{\text{motion}}$ jointly regresses a 13-dimensional optical-flow descriptor and classifies its motion class, and $\mathcal{L}_{\text{drift}}$ is an ℓ_2 anchor to the initialisation $\theta^{(0)}$ that bounds how far the adapted encoder may travel from the pretrained solution. Each term has a distinct role:

$\mathcal{L}_{\text{JEPA}}$	: future-latent prediction,
$\mathcal{L}_{\text{probe}}$	: checkpoint selection signal,
$\mathcal{L}_{\text{motion}}$	: motion grounding,
$\mathcal{L}_{\text{drift}}$	: catastrophic-forgetting control.

Predictor Surgery

FactorJEPA is initialised from a pretrained V-JEPA model; the online encoder, momentum target encoder, target construction and masking policy are all retained. Because only the top- K blocks and the predictor are updated, and because $\mathcal{L}_{\text{drift}}$ penalises displacement from $\theta^{(0)}$, adaptation is deliberately *local*: it changes how the predictor is trained, not what it is made of. We refer to this as **predictor surgery**, and evaluate two ablations of it: **FactorJEPA-RAW**, which applies the identical schedule and budget to raw clips with no factor views, and a **no- D_I** variant, which omits the interaction stage. Exact stage boundaries, mixture weights, unfreeze fractions, optimiser settings and measured GPU-hours are reported in Appendix C.

Experiments & Evaluation

We evaluate whether FactorJEPA improves the predictive structure of V-JEPA rather than only its downstream semantics. Our analysis centers on four diagnostics: **Future-frame**

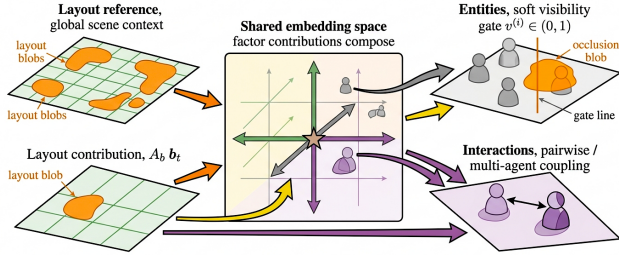


Figure 4: **Factorized predictive channels.** Context is decomposed into *layout*, visibility-gated *agent*, and sparse *interaction* coordinates, then composed in the future JEPA embedding. Separate subspaces preserve spatial structure, partially observed agents, and multi-agent coupling while limiting cross-factor leakage.

L1, Causal future-block L1, Mask-ratio slope, and Motion cosine. Together, they measure future-latent accuracy, causal future-block prediction, robustness to missing visual evidence, and linearly accessible motion information.

Evaluator Independence and Held-out Split

To prevent agreement with the pseudo-label generator from being mistaken for improved world modeling, we separate training-target construction from evaluation. factor targets from the Grounding DINO + SAM 3 pipeline are constructed only on the training split:

$$\mathcal{T}_{\text{train}} = \Psi_{\text{struct}}(\mathcal{F}_{\text{factor}}(x)), \quad x \in \mathcal{D}_{\text{train}}.$$

Primary evaluation is performed on a held-out split $\mathcal{D}_{\text{eval}}$ that is disjoint from the training split:

$$\mathcal{D}_{\text{train}} \cap \mathcal{D}_{\text{eval}} = \emptyset.$$

This held-out split is not used for training, model selection, threshold selection, or hyperparameter tuning. Future-frame L1 and Mask-ratio slope use only the frozen V-JEPA target encoder; Motion cosine uses an independently frozen motion estimator; and RGB Agent F1 uses a frozen detector that supplies neither FactorJEPA supervision nor factor-target preprocessing. Thus, no headline metric reuses FactorJEPA’s pseudo-label generator as its evaluator.

Protocol and Comparisons

We evaluate V-JEPA 2.1 at two scales: ViT-G with approximately 2B parameters and ViT-g with approximately 1B parameters. The comparison spans the frozen backbone, conventional full and parameter-efficient fine-tuning, LoRA, DoRA, Auto-RGN, **FactorJEPA-RAW**, and full **FactorJEPA**. FactorJEPA-RAW applies the factorized predictor to raw clips without factor-target supervision, separating architectural gains from those introduced by the factor curriculum.

We additionally evaluate **FactorJEPA-RAW**, which applies the factorized predictor architecture to the same raw

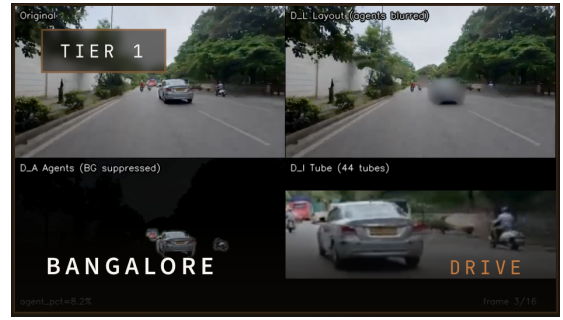


Figure 5: **DINO-based automatic factor-target construction.** From each input clip, the pipeline derives complementary supervision for *layout* D_L by suppressing foreground agents, *agents* D_A by suppressing background context, and *interactions* D_I through temporally associated agent tubes. These targets semantically anchor FactorJEPA’s predictive channels.

clips without factor-target supervision. The resulting attribution chain,

$$\underbrace{\{\text{LoRA, DoRA, Auto-RGN}\}}_{\text{generic adaptation}} \rightarrow \underbrace{\text{FactorJEPA-RAW}}_{\text{factorized architecture}} \rightarrow \underbrace{\text{FactorJEPA}}_{\text{architecture} + \text{factor curriculum}}$$

separates gains due to parameter adaptation, predictor structure, and structured supervision. Results in Section therefore test a precise claim: whether explicit **layout-agent-interaction structure** improves future prediction beyond the adaptation capacity of a monolithic JEPA predictor under a matched protocol.

All methods use the same clips, masking policy, evaluation split, and metric implementations. We report per-encoder test results with 95% BCa bootstrap confidence intervals. In the exported evaluation figures, *surgery* denotes FactorJEPA and *surgery-raw* denotes FactorJEPA-RAW.

Table 2: **Training and evaluation provenance.** No headline evaluator supplies FactorJEPA supervision.

Signal	Source	Factor-target source?
Factor supervision	Grounding DINO + SAM 3	Yes
Future latent	Frozen target encoder	No
Motion descriptor	Frozen motion estimator	No
RGB Agent F1	Independent detector	No

Four Diagnostics of Predictive Structure

Future-frame L1 ($\downarrow$). Normalized L1 distance between predicted and target future embeddings; lower values indicate **more accurate future-latent prediction**.

Causal future-block L1 ($\downarrow$). We split each clip’s tokens into a past temporal half and a future temporal half. Conditioning only on the past half, the predictor forecasts the latent representations of the future half, which we compare to the frozen target encoder’s latents on that same half:

$$\hat{Y}_{\text{fut}} = \hat{Y}(x_{\geq T/2} | x_{< T/2}), \quad Y_{\text{fut}}^* = \text{sg}[\bar{f}_{\hat{\theta}}(x)_{\geq T/2}].$$

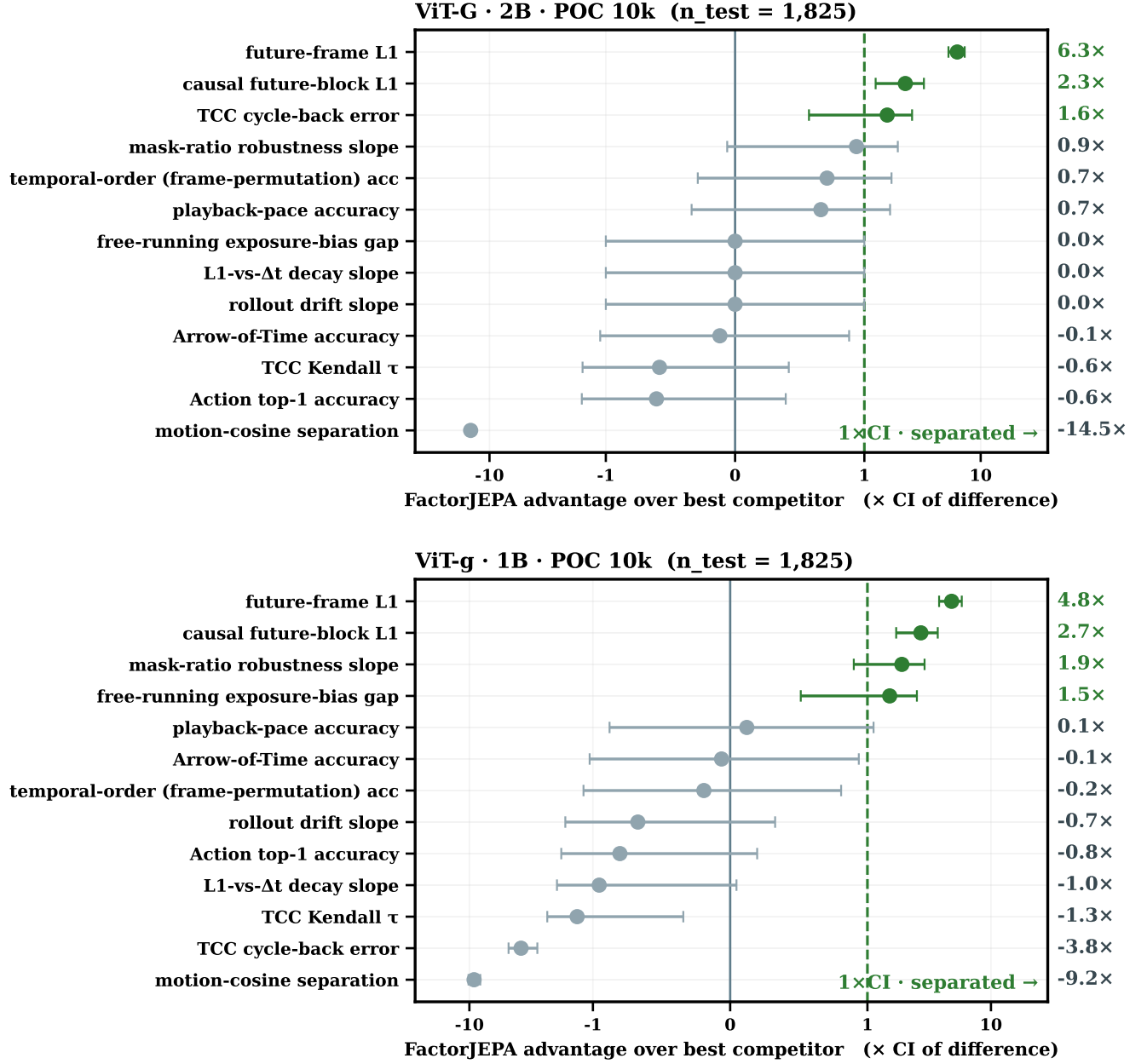


Figure 6: *FactorJEPA separates from the best fine-tuning competitor*. Points report FactorJEPA’s signed advantage over the *strongest* competitor for each metric, in units of the 95% confidence interval of the paired difference; the dashed line marks separation at $1\times$. On the 10k screening set at both 2B and 1B, FactorJEPA separates over every competitor on future-frame and causal future-block prediction, and on mask-ratio robustness at the 1B scale. Values measure confidence-interval separation, not multiplicative performance gains.

$$\text{CausalFutureL1} = \frac{1}{|\mathcal{D}|} \sum_{x \in \mathcal{D}} \frac{\|\hat{Y}_{\text{fut}} - Y_{\text{fut}}^*\|_1}{\epsilon + \|Y_{\text{fut}}^*\|_1}.$$

The temporal split is fixed and strictly causal (the past half predicts the future half) and uses no factor annotations; lower values indicate more accurate causal future-block prediction.

Mask-ratio slope ($\downarrow$). The increase in prediction error as visual evidence is progressively masked; a smaller slope indicates *greater robustness under partial observability*.

Motion cosine ($\uparrow$). Separation between same-motion and different-motion clips in pooled feature space, measured as intra-class minus inter-class cosine similarity; higher values indicate *motion is more directly encoded*.

Decisive Gains over the Frozen World Model

Figure 6 reports FactorJEPA’s advantage over the frozen backbone in confidence-interval widths; values beyond $1\times$ indicate statistical separation. FactorJEPA clears this threshold on **all four headline diagnostics at both scales**.

At 2B/1B, separation reaches $47.0\times/27.2\times$ for Future-frame L1, $28.7\times/19.0\times$ for Causal future-block L1, $26.8\times/25.6\times$ for Motion cosine, and $22.3\times/15.7\times$ for Mask-ratio slope. These are confidence-relative effect strengths, not multiplicative performance gains, and remain decisive after scaling from 2B to 1B.

Figure 7 isolates the two metrics on which FactorJEPA separates from *every* baseline at both scales—Future-frame L1 and Causal future-block L1. The broader picture is a **structured, not universal, gain**: across the full fifteen-diagnostic scorecard (Figure 12) FactorJEPA leads on predictive and masking diagnostics, while frozen or conventionally tuned models retain advantages on motion linearity and several frame-timing and temporal-coherence measures.

Prediction–Motion Trade-off

Against the strongest non-FactorJEPA adaptation, FactorJEPA improves **Future-frame L1** by 1.7%/1.4% at 2B/1B ($6.3\times/4.8\times$ CI widths), **Causal future-block L1** by 0.9%/1.1% ($2.3\times/2.7\times$), and **Mask-ratio slope** by 1.1%/2.9%, although the 2B masking result remains marginal.

The trade-off is **Motion cosine**. FactorJEPA improves motion accessibility over the frozen backbone, but trails the strongest full-fine-tuning arm by 44.9%/33.4% at 2B/1B. Thus, factorization favors **accurate, causally structured, and occlusion-robust prediction** over maximally linear motion readout, a trade-off that replicates across scales.

Cross-Scale Replication

We next ask whether the complete ranking of adaptation methods survives the transition from the 2B backbone to the approximately half-cost 1B backbone. For each diagnostic k we compute the Spearman rank correlation ρ_k between the per-method scores at the two scales.

The rankings are highly stable. Causal future-block L1 replicates most strongly ($\rho = 0.978$; Figure 8), followed by Motion cosine ($\rho = 0.952$), Future-frame L1 ($\rho = 0.938$), and Mask-ratio slope ($\rho = 0.895$)—the four headline diagnostics and the four strongest cross-scale correlations in the evaluation. Twelve of the fifteen diagnostics retain broadly consistent rankings, while Temporal-order, Teacher-free gap, and Rollout drift do not transfer reliably; the complete fifteen-panel comparison, with per-diagnostic leave-one-method robustness ranges, is in Figure 13.

The replication result is important for two reasons. First, it shows that both the predictive gains and the motion trade-off are stable properties of the method ordering rather than artifacts of one backbone. Second, it establishes the 1B model as a faithful lower-cost proxy for screening adaptation choices before full-scale 2B training.

Evaluation takeaway. Across both scales, FactorJEPA delivers (i) **more accurate future-latent prediction**, (ii) **stronger causal future-block structure**, and (iii) **greater robustness under reduced visual evidence**. It also makes motion substantially more accessible than in the frozen world model, while conceding motion linearity to the strongest generic fine-tuning arm. These effects replicate with $\rho = 0.895\text{--}0.978$, showing that explicit factorization changes not only prediction error, but the structure and accessibility of the information used to forecast the future.

Ablations and Limitations

We isolate each ingredient with matched controls on the same held-out split with paired BCa bootstrap; below, *separates* means the oriented 95% confidence interval excludes zero.

The gain comes from predictor surgery, not the factor targets. FactorJEPA-RAW applies the identical three-stage schedule and unfreeze budget to raw clips with no factor-target supervision. Against it, full FactorJEPA does not improve future-frame prediction: at both 2B and 1B the no-target control separates in its favour. The ingredient that closes the frozen-to-adapted gap is the surgical adaptation of the predictor, not the factor-view supervision layered on top of it.

Updating the encoder, not only the head, drives the predictive wins. Freezing the encoder and adapting only the attentive head loses to the encoder-updating variant on future-frame, causal future-block, and mask-ratio (each separated), while winning back motion-cosine, rollout, and exposure-bias. Touching the top encoder blocks is what buys the predictive-structure improvements, at the cost of some temporal and motion diagnostics.

The interaction stage adds no measurable value. Removing the interaction stage (no- D_I) leaves every predictive diagnostic statistically unchanged and separates in favour of the shorter schedule on motion-cosine and future-frame. We keep the stage for completeness but do not claim it as a source of gain.

Weight merging hurts. A WiSE-FT interpolation between the surgical and pretrained weights (mixing fractions 0.3, 0.5, 0.7) loses to the un-merged model on future-frame and motion-cosine at both scales. We report it as a negative result.

Limitations. Three controls we consider important were not executed and bound our claims. First, a compute-matched control that continues plain pretraining for the same number of steps as the surgery, isolating whether the effect is simply more optimisation, is declared in our configuration but was not run; we therefore cannot fully separate the surgical schedule from additional training. Second, head-only pretraining and the continual-learning baselines (CaSSLe, EWC) were trained but not evaluated on the diagnostic suite. Third, the method is a precondition on the base checkpoint: on the earlier V-JEPA 2.0 backbone the same recipe does not improve over the frozen encoder, so the gains reported here are specific to V-JEPA 2.1. Finally, at the 10k screening scale factorization concedes maximally linear motion readout to full

fine-tuning; this trade-off reverses only at full (86,831-video) scale.

Conclusion

We studied world modeling in **populous, crowded, and chaotic Global South cities**, where heterogeneity, occlusion, and mixed-traffic negotiation challenge conventional JEPA predictors. We introduced **DENSEWORLD: 276 hours** of drive-through, walk-through, and aerial video collected across **22 cities**. Building on this benchmark, we proposed **FactorJEPA**, which makes *world structure a first-class predictive primitive* through *predictor surgery*: a three-stage curriculum that adapts the predictor over automatically separated *layout*, *agent*, and *interaction* views. Across 2B and 1B backbones, FactorJEPA delivers (i) *more accurate future-latent prediction*, (ii) *stronger causal future-block structure*, and (iii) *greater robustness under reduced visual evidence*, with method rankings reproducing across scale at $\rho = 0.895$ to 0.978 . The results also expose a trade-off between structured prediction and maximally linear motion readout at the 10k screening scale, which reverses at full scale (86,831 factor-annotated videos). Together, these contributions move JEPA world models from *monolithic predictors* toward *structured, factor-aware models* of complex real-world dynamics.

Winner scorecard: FactorJEPA vs. all baselines (POC 10k ($n_{\text{test}} = 1,825$) · 95% BCa CI)

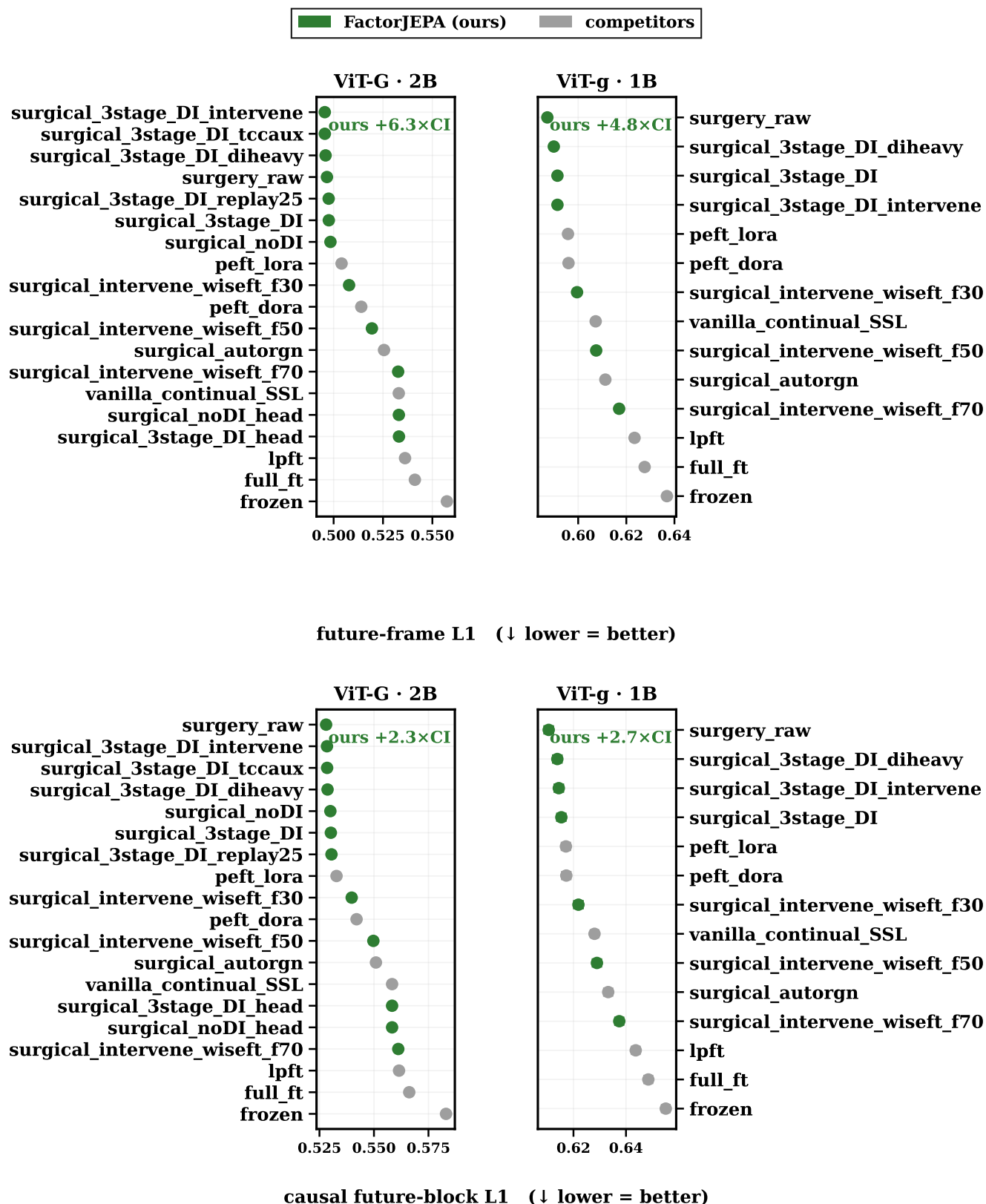


Figure 7: *FactorJEPA wins the two scale-robust predictive metrics against every baseline.* Per-arm held-out score $\pm$ 95% BCa CI (dot, not bar-from-zero, so the small but significant margins stay visible; FactorJEPA green, competitors grey, best-at-top) on the two metrics that separate over the best competitor at *both* scales—Future-frame L1 (top), Causal future-block L1 (bottom), at 2B (left) and 1B (right). FactorJEPA holds the winning end of every panel while the frozen encoder trails; the annotated margin is the best-ours advantage in CI widths. The complete fifteen-diagnostic scorecard is in Figure 12.

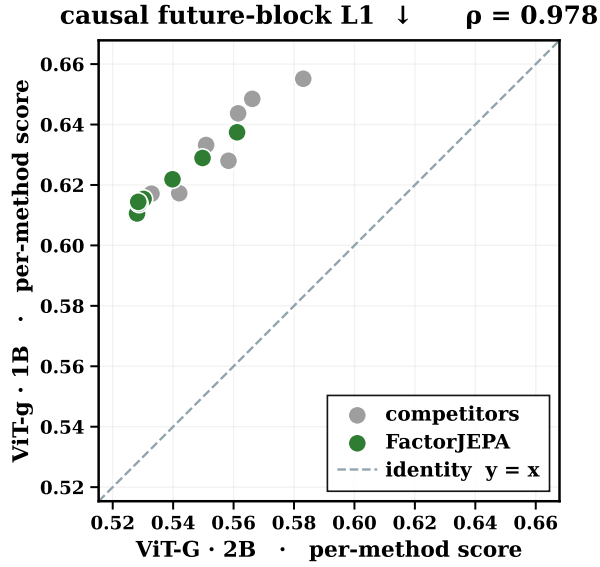


Figure 8: *Method rankings survive the 2B→1B backbone change.* Per-method scores at the ViT-G 2B and ViT-g 1B scales for *Causal future-block L1*—the strongest-replicating of the fifteen diagnostics (Spearman $\rho = 0.978$). Each point is one adaptation method (FactorJEPA in green, competitors in grey); the near-perfect rank agreement establishes the 1B backbone as a faithful low-cost proxy for screening adaptation choices before full-scale 2B training. Points sit above the identity line because the smaller backbone has uniformly higher causal error; the *ranking*, not the absolute level, is what replicates. The complete fifteen-panel replication grid is in Figure 13.

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Appendix

This appendix provides the complete evidence and implementation record underlying the main paper. Its organization follows the scientific progression of the study. Appendices A–C establish the provenance, distinctiveness, and reproducibility of DENSEWORLD, its factor targets, and the executed training protocols. Appendices D–F then examine the central methodological claims: which components produce the observed gains, whether the evaluation is statistically reliable, and whether the learned layout, agent, and interaction channels are functionally distinct. Appendix G evaluates the temporal and downstream consequences of the prediction–motion trade-off.

The appendix distinguishes primary confirmatory evidence from diagnostic and robustness analyses. Unless otherwise stated, comparisons use identical evaluation clips, paired resampling units, and the same independently defined audit signals. Factor separation is interpreted operationally at the level of predictive channels rather than as coordinate-level identifiability or complete statistical independence. Causal future-block evaluation measures predictive accuracy on the future temporal half, while cross-scale replication refers specifically to stability between the 1B and 2B V-JEPA 2.1 backbones.

Appendix index.

- **Appendix A: DENSEWORLD Construction, Splits, and Responsible Use**
 - **A.1 Acquisition protocol and capture modes**
Shared collection specification; geographic and environmental sampling; non-scripted scene activity; and drive-through, walk-through, and aerial capture.
 - **A.2 Source processing, clip construction, and provenance**
FFmpeg decoding; adaptive shot-boundary detection; model-ready clip construction; quality filtering; and

the retained city–session–source–shot–clip provenance hierarchy.

- **A.3 Partitioning and factor target provenance**

Source-grouped train–validation–test allocation (86,831 / 909 / 23,106 clips); partition independence; permitted information flow; and automatically derived training, validation, and teacher-relative test targets. No manually annotated semantic labels are used.

- **A.4 Corpus composition and coverage**

City and capture-mode distributions; environmental and temporal coverage; scene families; traffic composition; and the automated dynamic-agent taxonomy, including locally distinctive agents.

- **A.5 Privacy processing, responsible use, and research artifacts**

Automated face and registration-plate detection; temporal association and blurring; redaction quality control; governed research access; supporting code and aggregate statistics; intended and prohibited uses; and empirical scope and extensions.

- **Appendix B: Five-Axis Validation of the DENSEWORLD Regime**

- **B.1 Shared measurement protocol and automated taxonomy**

Common preprocessing, temporal sampling, valid-image regions, Grounding DINO + SAM 3 extraction, dynamic-agent taxonomy, and source-level aggregation across DENSEWORLD, BDD100K, and nuScenes.

- **B.2 Definition of the five regime axes**

Formal definitions of agent count density, agent occupancy, occlusion pressure, interaction pressure, and agent heterogeneity, including normalization, aggregation units, and edge-case handling.

- **B.3 Matched cross-dataset comparison**

Road-level scene matching by capture configuration, scene type, illumination, weather, image resolution, field of view, and clip duration. Walk-through and aerial DENSEWORLD clips are analyzed separately from the primary driving-benchmark comparison.

- **B.4 Comparative results, uncertainty, and sensitivity**

Five-axis estimates, scene-stratified effects, source-video-clustered confidence intervals, and hierarchical resampling over cities or sequences. Sensitivity is evaluated over detection threshold, minimum agent size, valid-image normalization, and interaction parameters. Results are interpreted as evidence for an Indian urban realization of the broader DENSEWORLD regime, not as an exhaustive characterization of Global-South mobility.

- **Appendix C: Segmentation and Training**

- **C.1 Frozen Teacher and Factor-Target Construction**

Grounding DINO + SAM 3 checkpoint, preprocessing, feature extraction, region and mask construction, confidence filtering, temporal association, and the exact

definitions of the layout, agent, visibility, and interaction targets T_L, T_A, T_V, T_I . This subsection also specifies target dimensions, interaction-pair construction, reliability weights, exclusion rules, and retained target coverage.

- **C.2 FactorJEPA Architecture and Prediction Protocol**

V-JEPA 2.1 backbone scales, frozen and trainable components, predictor depth, factor projections, subspace ranks, visibility gating, masking policy, context duration, prediction horizon, and factor-to-predictor information flow.

- **C.3 Optimization, Model Selection, and Executed Algorithm**

Objective terms, loss weights, optimizer, learning-rate schedule, momentum-target update, batch size, gradient clipping, training steps, random seeds, validation criteria, checkpoint selection, and end-to-end training pseudocode.

- **C.4 Baseline Matching and Attribution Controls**

Full fine-tuning, LoRA, DoRA, Auto-RGN, FactorJEPA-RAW, and FactorJEPA configurations; trainable-parameter matching; shared data, masks, objectives, training steps, and evaluation protocol; and the controls separating generic adaptation, factorized architecture, and factor-target supervision.

- **C.5 Evaluation Provenance and Resource Accounting**

Separation of training targets, validation signals, and frozen test evaluators; software and hardware configuration; trainable and total parameters; GPU-hours; peak memory; throughput; inference latency; and the reproducibility record for every reported experiment.

- **Appendix D: Attribution and Component Ablations**

- **D.1 Attribution design and experimental contract**

Defines the architectural and supervisory interventions, the parameter-matching protocol, the executed objectives, and the statistical estimands used throughout the appendix.

- **D.2 Architecture $\times$ structured-supervision factorial**

Compares matched monolithic and factorized predictors with and without automatically derived structured supervision, isolating architecture, supervision, and their interaction.

- **D.3 Factor pathways and objective ablations**

Tests factor-view ablations across the layout, agent, and interaction views through controlled one-factor removals.

- **D.4 Falsification controls and sensitivity**

Evaluates temporally misaligned and clip-shuffled teacher targets, reliability weighting, factor ranks, and loss coefficients to distinguish structured information from generic regularization.

- **D.5 Attribution synthesis**

Consolidates the factorial contrasts and component ablations, reporting which mechanisms account for each

prediction, visibility, interaction, motion, and RGB result.

- **Appendix E: Metrics, Clustered Inference, and Robustness**

- **E.1 Evaluation contract and metric registry**

Defines every reported diagnostic by formula, direction, evaluation unit, aggregation rule, frozen evaluator, provenance, and primary or exploratory status.

- **E.2 Primary predictive diagnostics**

Specifies Future-frame L1, causal future-block L1, Mask-ratio slope, and Motion cosine, including token aggregation, normalization, probe training, and horizon or masking support.

- **E.3 Secondary semantic and temporal diagnostics**

Defines action, taxonomy, rollout, temporal-order, arrow-of-time, playback-pace, temporal-correspondence, and exposure-bias diagnostics under a common aggregation protocol.

- **E.4 Paired clustered inference and multiplicity**

Defines the city-source-video estimand, hierarchical paired bootstrap, treatment of training seeds, mixed-effects sensitivity, and multiple-comparison control.

- **E.5 Metric sensitivity and complete numerical results**

Reports horizon and masking curves, normalization sensitivity, cross-scale ranking robustness, leave-one-method and leave-one-family analyses, and the complete ViT-G and ViT-g numerical scorecards.

DENSEWORLD Construction, Splits, and Responsible Use

DENSEWORLD comprises approximately **276 hours** of drive-through, walk-through, and aerial video acquired across **22 Indian cities**. It targets predictive modeling under high agent density, heterogeneous traffic, persistent occlusion, soft spatial boundaries, and frequent local interaction. DENSEWORLD contains *no manually annotated semantic labels*; all layout, agent, visibility, and interaction targets are produced automatically using a fixed Grounding DINO + SAM 3 pipeline.

Acquisition Protocol and Capture Modes

DENSEWORLD was assembled from **714 long-form YouTube videos** (480p) selected under a **shared coverage specification**. The specification defined coverage targets over geographic sites, scene types, times of day, weather, visibility, road structure, and infrastructure conditions. It controlled *where and how* videos were acquired without scripting traffic, pedestrian, or animal behavior.

The collection uses three capture modes. **Drive-through** recordings emphasize road-level ego-motion, mixed traffic, and near-field interactions. **Walk-through** recordings capture pedestrian-scale navigation, shared spaces, markets, commercial corridors, and transit areas. **Aerial** recordings provide broader context for crowd flow, junction organization, traffic structure, and interaction topology. City, collection session, capture mode, and source timestamps are retained as acquisition metadata.

Source Processing, Clip Construction, and Provenance

Source videos are decoded using FFmpeg¹ while retaining frame timestamps and source identifiers. Shot boundaries are detected using the PySceneDetect AdaptiveDetector², which normalizes adjacent-frame changes in hue, saturation, and luminance by a rolling temporal average. This reduces false boundaries caused by rapid ego-motion and camera shake.

We use a fixed corpus-wide configuration: `adaptive_threshold= 3.0`, `min_content_val= 15.0`, `window_width= 2`, and `min_scene_len= 15` frames. For a source video $v = \{x_t^{(v)}\}_{t=1}^{T_v}$, the detector returns ordered boundaries

$$\mathcal{B}_v = \left\{ 1 = b_0^{(v)} < b_1^{(v)} < \dots < b_{J_v}^{(v)} = T_v \right\},$$

which partition the source into candidate visually continuous shots. Shots shorter than the minimum usable duration are excluded; retained shots yield model-ready clips of approximately 6–13 seconds. No clip crosses a detected shot boundary.

The retained provenance hierarchy is

city $\rightarrow$ collection session $\rightarrow$ source video $\rightarrow$ shot $\rightarrow$ clip

¹<https://ffmpeg.org/>

²<https://www.scenesdetect.com/docs/latest/api/detectors.html>

For every clip c , we store

$$\pi(c) = (i_{\text{city}}, i_{\text{session}}, i_{\text{source}}, i_{\text{shot}}, m_{\text{capture}}, t_{\text{start}}, t_{\text{end}}).$$

Thus, the clip is the **computational input**, while its source, session, and city identifiers preserve higher-level geographic and temporal dependence.

Partitioning and Factor Target Provenance

Source videos are assigned as indivisible groups using an **source-grouped train/validation/test allocation** (86,831 / 909 / 23,106 clips; 4,841 clips without factor targets excluded), stratified by city and capture mode. Split assignment precedes clip extraction, so every shot and clip derived from one source video inherits the same partition. Let $\mathcal{V}_s$ denote the source-video set assigned to partition s . The executed grouping satisfies

$$\mathcal{V}_{\text{train}} \cap \mathcal{V}_{\text{val}} = \mathcal{V}_{\text{train}} \cap \mathcal{V}_{\text{test}} = \mathcal{V}_{\text{val}} \cap \mathcal{V}_{\text{test}} = \emptyset.$$

DENSEWORLD does not use manually annotated factor labels. For each partition, automated structural targets are generated by the fixed Grounding DINO + SAM 3 pipeline:

$$\mathcal{T}_s^{\text{DINO}} = \{ \Psi_{\text{struct}}(F_{\text{factor}}(x)) : x \in \mathcal{D}_s \}, \\ s \in \{\text{train, val, test}\}.$$

Training targets may provide factor supervision; validation targets may support prespecified selection and calibration; and test targets are used only for held-out *teacher-relative* factor diagnostics. Test-derived targets, thresholds, and metrics never influence gradient updates or model selection.

Corpus Composition and Coverage

Geographic and capture-mode composition. The collection spans metropolitan and additional urban sites across India. These groups are **collection strata**, not claims of an official administrative taxonomy or within-stratum homogeneity. Figure 9 reports the contribution of drive-through, walk-through, and aerial clips for each city. The city-by-mode distribution is observational and intentionally heterogeneous; bar length describes corpus composition, not evaluation weight or statistical independence.

Environmental and agent coverage. DENSEWORLD includes commercial, residential, transit, heritage, coastal, and high-density street environments, spanning markets, junctions, flyovers, highways, promenades, bazaars, ghats, beaches, and shared roads. It also captures variation in illumination, weather, traffic composition, crowd density, road geometry, surface quality, encroachment, and pedestrian-vehicle separation.

The dynamic-agent taxonomy extends beyond passenger vehicles and pedestrians to include auto-rickshaws, cycle rickshaws, delivery riders, multi-rider two-wheelers, cyclists, mobile vendors, animal-drawn vehicles, and independently moving animals. Figure 10 illustrates category semantics; it is neither a frequency-proportional sample nor manually annotated evidence. Quantitative prevalence and heterogeneity statistics derived from the automated pipeline are reported as *automatically estimated* quantities.

Table 3: **Composition, information flow, and independence constraints of the DENSEWORLD partitions.** Allocation is performed over source-video groups. Durations are approximate because grouped recordings vary in length and usable-shot yield. Grounding DINO + SAM 3 test measurements are reported as teacher-relative; headline prediction, motion, and RGB metrics use fixed evaluators that do not provide FactorJEPA’s factor-supervision targets.

Partition	Share	Duration	Gradient updates	Model selection	Threshold calibration	Signal exposed	Isolation constraint
Training	80%	≈ 800 h	Yes	No	No	JEPA prediction targets and automatically derived layout, agent, and interaction targets.	Source-video disjoint from validation and test; all descendants of a source remain grouped.
Validation	10%	≈ 100 h	No	Yes	Yes	Prespecified validation metrics and automatically derived targets used only for selection and calibration.	Source-video disjoint from training and test; never used for gradient optimization.
Test	10%	≈ 100 h	No	No	No	Frozen headline evaluators and automatically derived targets used only for explicitly teacher-relative diagnostics.	Source-video disjoint from training and validation; all configurations are frozen before test access.
Total	100%	≈ 1,000 h	–	–	–	–	–

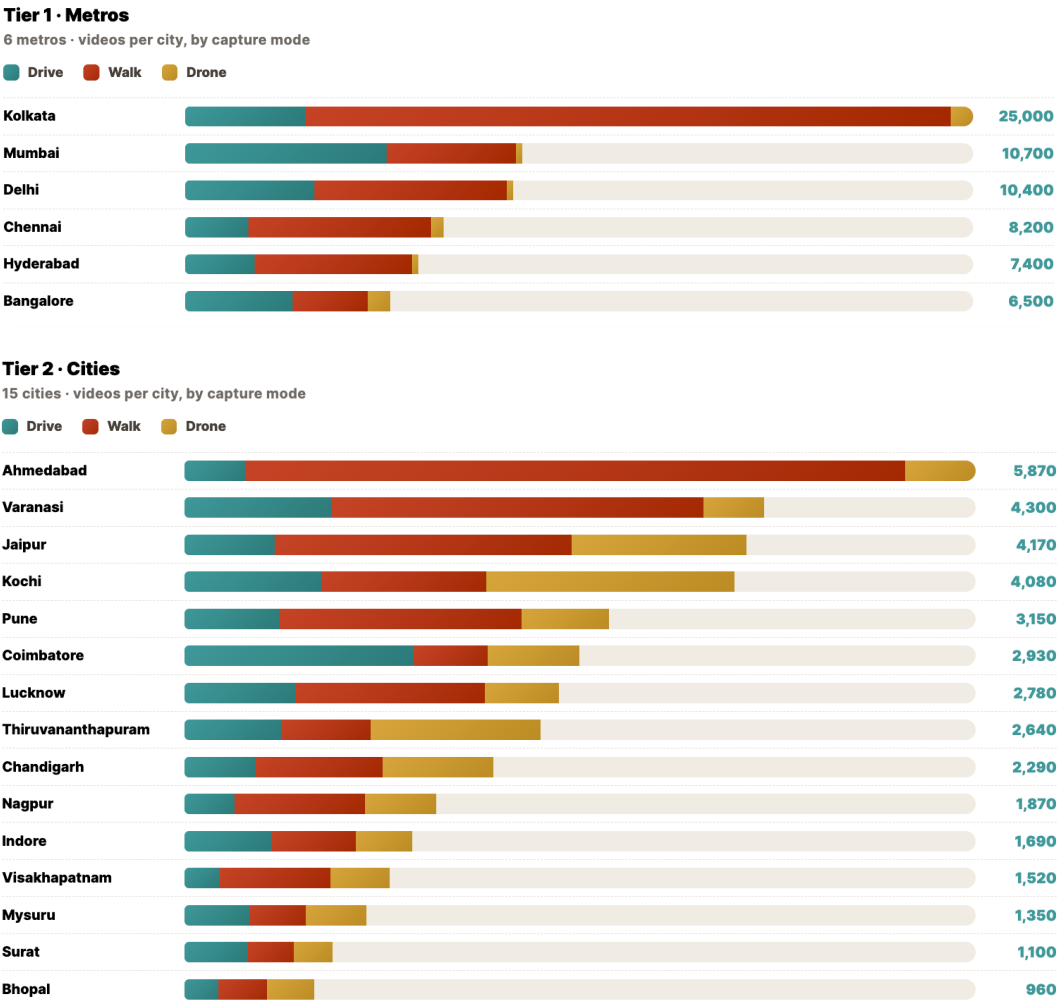


Figure 9: **City-level composition by capture mode.** The upper panel shows metropolitan collection sites and the lower panel shows additional urban sites. Each bar decomposes a city’s model-ready clips into drive-through, walk-through, and aerial capture. The value at the right gives the city total; both panels use a common horizontal scale.



Figure 10: **Illustrative DENSEWORLD dynamic-agent taxonomy.** The examples communicate the semantic scope of the automated taxonomy, including conventional transport, intermediate mobility, mobile vendors, animal-drawn transport, and independently moving animals. They are illustrative category renderings rather than frequency-proportional dataset samples.

Privacy Processing, Responsible Use, and Research Artifacts

Privacy processing. Before downstream use, clips pass through an automated privacy pipeline. Faces are localized with SCRFD³, and vehicle-registration plates are localized with a custom-trained YOLO detector⁴. Detections are associated temporally using ByteTrack⁵, expanded by a fixed spatial safety margin, and redacted using OpenCV GaussianBlur⁶.

The policy covers frontal, profile, small, and partially occluded faces, detectable faces in reflections, and readable or partially readable vehicle-registration plates. Let $\mathcal{R}_t^{\text{face}}$ and $\mathcal{R}_t^{\text{plate}}$ denote the temporally associated and expanded regions in frame x_t . The processed frame is

$$\tilde{x}_t = \mathcal{B}_\sigma \left(x_t, \mathcal{R}_t^{\text{face}} \cup \mathcal{R}_t^{\text{plate}} \right),$$

³<https://github.com/deepinsight/insightface/tree/master/detection/scrfd>

⁴<https://docs.ultralytics.com/tasks/detect/>

⁵<https://github.com/FoundationVision/ByteTrack>

⁶https://docs.opencv.org/4.x/d4/d86/group__imgproc__filter.html

where $\mathcal{B}_\sigma$ applies region-scaled Gaussian filtering. Automated quality control checks track discontinuities, unmatched detections, frame-boundary truncation, and processing failures. Segments with incomplete or failed redaction are excluded. Only $\tilde{x}_t$ enters target construction, training, evaluation, or external illustration.

Responsible use and research artifacts. DENSEWORLD is intended for predictive world modeling, video representation learning, multi-agent forecasting, and robustness under partial observability. It is not intended for person identification, persistent individual tracking, sensitive-attribute inference, or surveillance.

The complete corpus is maintained within the **governed research environment** used for the reported experiments. Supporting artifacts include the FactorJEPa implementation, preprocessing configuration, factor target-construction pipeline, metric definitions, automated taxonomy, aggregate corpus statistics, and representative privacy-filtered material, subject to applicable rights and privacy review.

Five-Axis Validation of the DENSEWORLD Regime

We ask whether DENSEWORLD remains quantitatively distinct from BDD100K (Yu et al. 2020) and nuScenes (Caesar et al. 2020) after controlling for observable differences in capture and scene composition. We characterize this regime along five axes: **agent count density**, **agent occupancy**, **visibility pressure**, **interaction pressure**, and **agent heterogeneity**.

The comparison is **outcome-blind**, **scene-matched**, and **dependence-aware**. All datasets are processed using the same frozen Grounding DINO + SAM 3 measurement pipeline; none of the five axes enters the matching procedure; and uncertainty is clustered above the frame and clip levels. The resulting quantities are therefore interpreted as *teacher-relative dataset statistics*, not manually annotated ground truth or causal effects of dataset membership.

Shared Measurement Protocol and Automated Taxonomy

All three datasets are processed through the common sequence

video $\rightarrow$ sampled frames $\rightarrow$ Grounding DINO + SAM 3 detections $\rightarrow$ structural pseudo-annotations $\rightarrow$ five-axis statistics

For frame $x_t^{(d)}$ from dataset $d \in \{\text{DW}, \text{BDD}, \text{NS}\}$, the frozen Grounding DINO + SAM 3 pipeline (Carion et al. 2025) and structural postprocessor produce

$$\begin{aligned} \mathcal{P}_t^{(d)} &= \Psi_{\text{struct}} \left(F_{\text{factor}}(x_t^{(d)}) \right) \\ &= \left\{ \left(\hat{c}_{i,t}, \hat{s}_{i,t}, \hat{B}_{i,t}, \hat{M}_{i,t}, \hat{\ell}_{i,t} \right) \right\}_{i \in \mathcal{I}_t^{(d)}}. \end{aligned}$$

Here, $\mathcal{I}_t^{(d)}$ indexes candidate instances; $\hat{c}_{i,t}$, $\hat{s}_{i,t}$, $\hat{B}_{i,t}$, and $\hat{M}_{i,t}$ denote the predicted class, confidence, bounding box, and visible instance mask; and $\hat{\ell}_{i,t}$ is the temporally associated track identifier.

SAM 3 supplies text-prompted instance masks directly, and Ψ_{struct} denotes the executed frozen sequence of proposal extraction, class assignment, mask construction, duplicate suppression, and temporal association. Its component architectures, checkpoint identifiers, inference resolution, thresholds, suppression rule, and association parameters are fixed before comparison and applied identically to all datasets.

Let Ω_t denote the valid, unpadded image region. The retained agent set is

$$\mathcal{A}_t^{(d)} = \left\{ i \in \mathcal{I}_t^{(d)} : \hat{s}_{i,t} \geq \tau_{\text{det}}, \frac{|\hat{M}_{i,t} \cap \Omega_t|}{|\Omega_t|} \geq a_{\min} \right\}.$$

The same temporal stride, resizing rule, confidence threshold, minimum-area criterion, and temporal-association configuration are used for DENSEWORLD, BDD100K, and nuScenes. Predictions that cannot be mapped unambiguously are excluded rather than reassigned using dataset-specific heuristics.

Predicted classes are mapped to seven shared superclasses:

- (i) **pedestrian**
- (ii) **pedal cycle**
- (iii) **powered two-wheeler**
- (iv) **light passenger vehicle**
- (v) **heavy or public vehicle**
- (vi) **intermediate or non-motorized transport**
- (vii) **animal or animal-drawn transport**

Native BDD100K and nuScenes annotations are not used to compute, calibrate, or select any primary statistic. To expose possible teacher-induced domain effects, we report the retained-instance rate, unresolved-class rate, confidence distribution, small-instance exclusion rate, and track-fragmentation rate separately for each dataset. The primary analysis is additionally repeated over a prespecified grid of confidence, area, and association settings.

Definition of the Five Regime Axes

Suppressing the dataset superscript for clarity, let

$$N_t = |\mathcal{A}_t|, \quad \mathcal{T}_1 = \{t : N_t \geq 1\}, \quad \mathcal{T}_2 = \{t : N_t \geq 2\}.$$

All boxes and masks are clipped to Ω_t before measurement. Count and occupancy are evaluated over all sampled frames. Visibility pressure is evaluated over $\mathcal{T}_1$, whereas interaction and heterogeneity are evaluated over $\mathcal{T}_2$. We report the eligible-frame rate for every conditional statistic.

1. Agent count density.

The primary statistic is the mean number of retained dynamic agents per sampled frame:

$$D_{\text{count}} = \frac{1}{T} \sum_{t=1}^T N_t.$$

We additionally report valid-support-normalized count,

$$D_{\text{count}}^{\text{area}} = \frac{1}{T} \sum_{t=1}^T \frac{N_t}{|\Omega_t|/10^6},$$

as a sensitivity measure for cropping, padding, and inference resolution. It is not interpreted as physical agent density.

2. Agent occupancy.

Occupancy is the fraction of valid image support covered by the union of visible agent masks:

$$D_{\text{occ}} = \frac{1}{T} \sum_{t=1}^T \frac{\left| \bigcup_{i \in \mathcal{A}_t} (\hat{M}_{i,t} \cap \Omega_t) \right|}{|\Omega_t|}.$$

Taking the union prevents crowded or overlapping regions from being counted multiple times.

3. Visibility pressure.

Because no amodal annotations are available, this axis is defined as an automated *visibility-pressure proxy*. It combines three observable cues: substantial pairwise box

overlap, valid-image-boundary truncation, and short interior track disappearance followed by reappearance. For boxes B_i and B_j , define intersection over minimum area as

$$\text{IoM}(B_i, B_j) = \frac{|B_i \cap B_j|}{\min(|B_i|, |B_j|) + \epsilon}.$$

The agent-level visibility indicator is

$$z_{i,t} = \mathbf{1} \left[\max_{j \neq i} \text{IoM}(\hat{B}_{i,t}, \hat{B}_{j,t}) \geq \tau_{\text{ov}} \vee e_{i,t} = 1 \vee g_{i,t} = 1 \right]$$

where $e_{i,t}$ indicates boundary truncation and $g_{i,t}$ indicates an interior track gap followed by reappearance within the prespecified temporal window. Visibility pressure is

$$D_{\text{vis}} = \frac{1}{|\mathcal{T}_1|} \sum_{t \in \mathcal{T}_1} \frac{\sum_{i \in \mathcal{A}_t} z_{i,t}}{N_t}.$$

Overlap, boundary-truncation, and track-gap rates are also examined separately to verify that the composite is not dominated by one proxy.

4. Interaction pressure.

We construct a frame-level interaction graph $\mathcal{G}_t = (\mathcal{A}_t, \mathcal{E}_t)$. Let $\mathbf{p}_{i,t}$ denote the normalized image-plane center of agent i . Velocities $\tilde{\mathbf{u}}_{i,t}$ are computed after removing estimated global image motion. Define

$$\mathbf{r}_{ij,t} = \mathbf{p}_{j,t} - \mathbf{p}_{i,t}, \quad \mathbf{w}_{ij,t} = \tilde{\mathbf{u}}_{j,t} - \tilde{\mathbf{u}}_{i,t}.$$

For approaching pairs, the predicted time and distance at closest approach are

$$t_{ij,t}^* = -\frac{\mathbf{r}_{ij,t}^\top \mathbf{w}_{ij,t}}{\|\mathbf{w}_{ij,t}\|_2^2 + \epsilon},$$

$$d_{ij,t}^* = \|\mathbf{r}_{ij,t} + t_{ij,t}^* \mathbf{w}_{ij,t}\|_2.$$

An interaction edge is present when the pair is currently proximate or is predicted to approach within both the temporal and spatial thresholds:

$$(i, j) \in \mathcal{E}_t \iff \|\mathbf{r}_{ij,t}\|_2 \leq \rho \vee (0 < t_{ij,t}^* \leq \tau_t \wedge d_{ij,t}^* \leq \tau_d).$$

Interaction pressure is the conditional mean graph degree:

$$D_{\text{int}} = \frac{1}{|\mathcal{T}_2|} \sum_{t \in \mathcal{T}_2} \frac{2|\mathcal{E}_t|}{N_t}.$$

Per-pair edge density is retained as a sensitivity statistic to separate interaction structure from the mechanical effect of agent count.

5. Agent heterogeneity.

Let $N_{c,t}$ be the number of retained agents assigned to superclass $c \in \mathcal{C}$. The primary statistic is the probability that two agents drawn without replacement from the same frame belong to different superclasses:

$$h_t = 1 - \sum_{c \in \mathcal{C}} \frac{N_{c,t}(N_{c,t} - 1)}{N_t(N_t - 1)}, \quad t \in \mathcal{T}_2.$$

Dataset-level heterogeneity is

$$D_{\text{het}} = \frac{1}{|\mathcal{T}_2|} \sum_{t \in \mathcal{T}_2} h_t.$$

This pairwise definition lies in $[0, 1]$ and is less sensitive to small per-frame counts than plug-in entropy. Normalized entropy, class richness, and the effective number of classes are retained as secondary measures.

Matched Cross-Dataset Comparison

The primary analysis matches DENSEWORLD drive-through clips separately against road-level clips from BDD100K (Yu et al. 2020) and nuScenes (Caesar et al. 2020). Walk-through and aerial clips are summarized separately and do not enter the matched driving-benchmark comparison.

Matching is performed without access to the five axes or their component measurements. Clips are first assigned to common strata defined by capture mode, scene family, illumination, weather, and road context. These nuisance attributes are obtained using the same fixed metadata rules across datasets. Within each stratum, we perform 1:1 nearest-neighbor matching without replacement over standardized clip duration, valid image support, aspect ratio, resolution, and common camera metadata. Covariates unavailable consistently across all datasets are excluded from the primary matching model.

Let $\mathbf{z}_v$ denote the matching covariates for clip v . Post-matching balance is measured using the standardized mean difference

$$\text{SMD}(z) = \frac{\bar{z}_{\text{DW}} - \bar{z}_{\text{ref}}}{\sqrt{(s_{\text{DW}}^2 + s_{\text{ref}}^2)/2}}.$$

The primary analysis retains common-support strata satisfying

$$\max_{z \in \mathbf{z}} |\text{SMD}(z)| < 0.1.$$

We report the number of candidate and retained clips, independent source groups, unmatched fraction, and maximum absolute SMD before and after matching. Unmatched clips are excluded from the primary cross-dataset estimate and retained only in DENSEWORLD-specific coverage summaries.

For axis k , let $\mathcal{T}_{k,u}$ be the eligible sampled frames from matched clips belonging to source group u . The source-level summary is

$$\bar{m}_{k,u}^{(d)} = \frac{1}{|\mathcal{T}_{k,u}|} \sum_{t \in \mathcal{T}_{k,u}} m_{k,t}^{(d)}.$$

The matched-population estimate gives equal weight to every retained source group:

Table 4: **Scene-matched agent density and occupancy.** Count density is measured in agents per sampled frame; occupancy is the percentage of valid image area covered by the union of visible agent masks. $\Delta = \text{DW} - \text{reference}$, and R is the ratio of the two means. DENSEWORLD results are aggregated over K executed folds. The final row gives the unweighted mean across the six matched scene strata.

Matched scene stratum	Agent count density (agents/frame)				Agent occupancy (% valid area)			
	Matched reference	DENSEWORLD [K-fold]	Δ	R	Matched reference	DENSEWORLD [K-fold]	Δ_{pp}	R
Residential lane	1.8	3.3	+1.5	1.83×	2.0%	3.5%	+1.5	1.75×
Promenade	1.4	2.5	+1.1	1.79×	1.5%	2.8%	+1.3	1.87×
Market	4.6	12.4	+7.8	2.70×	4.2%	11.1%	+6.9	2.64×
Heritage / tourist	1.7	3.0	+1.3	1.76×	0.9%	1.4%	+0.5	1.56×
Flyover / underpass	2.6	5.0	+2.4	1.92×	8.8%	20.8%	+12.0	2.36×
Commercial	4.1	11.0	+6.9	2.68×	3.9%	9.0%	+5.1	2.31×
Equal-stratum mean	2.70	6.20	+3.50	2.30×	3.55%	8.10%	+4.55	2.28×

Interpretation. The ratios in the final row are ratios of the equal-stratum means, not means of the six stratum-specific ratios. These quantities characterize the measured operating regime; larger values do not denote conventional benchmark performance.

$$\hat{\mu}_{r,k}^{(d)} = \frac{1}{|\mathcal{U}_{d,r}|} \sum_{u \in \mathcal{U}_{d,r}} \bar{m}_{k,u}^{(d)},$$

where $r \in \{\text{BDD100K}, \text{nuScenes}\}$ identifies the reference-specific matched population. The DENSEWORLD estimate is therefore recomputed for its BDD100K-matched and nuScenes-matched subsets.

For each reference r , we report the absolute difference and ratio

$$\Delta_{r,k} = \hat{\mu}_{r,k}^{(\text{DW})} - \hat{\mu}_{r,k}^{(r)}, \quad R_{r,k} = \frac{\hat{\mu}_{r,k}^{(\text{DW})} + \epsilon}{\hat{\mu}_{r,k}^{(r)} + \epsilon}.$$

These contrasts describe separation under matched observable conditions; they are not interpreted causally.

Comparative Results, Uncertainty, and Sensitivity

The primary five-axis comparison is reported below. Entries follow the form $X [X, X]$: a fold-aggregated mean followed by its hierarchical 95% confidence interval. The bracketed value after **DENSEWORLD** denotes the number of executed folds.

Confidence intervals are estimated using a **hierarchical clustered bootstrap**. DENSEWORLD resamples cities and then source videos within cities; BDD100K resamples source videos; and nuScenes resamples collection logs and scenes. All clips from a sampled source group remain together, and matching is re-estimated inside every bootstrap replicate. We report 95% bias-corrected and accelerated (BCa) intervals for dataset means, absolute differences, and ratios. Holm correction is applied jointly to the ten primary contrasts: five axes against two reference datasets.

Robustness is evaluated over prespecified variations of τ_{det} , a_{min} , τ_{ov} , ρ , (τ_t, τ_d) , and temporal sampling rate. We also compare source and equal-stratum weighting, conditional and unconditional normalization, and alternative

common-support calipers. Leave-one-city-out analysis tests whether any individual DENSEWORLD city determines the reported effects.

Two automated negative controls test metric specificity. *Spatial permutation* preserves frame-level counts and class frequencies while disrupting proximity and closest-approach edges. *Temporal permutation* preserves frame-level count and occupancy while disrupting motion consistency and track-gap structure. Each metric must attenuate under its corresponding control without mechanically altering the axes that the control is intended to preserve.

An axis is interpreted as exhibiting stable regime separation only when its effect direction is preserved across the prespecified sensitivity grid, its clustered interval excludes zero, and the conclusion survives leave-one-city-out analysis. The resulting evidence is specific to the shared automated measurement system and matched observable conditions. It does not imply exhaustive geographic coverage, causal effects of dataset membership, or universal superiority over existing driving benchmarks.

Segmentations, and Training

This appendix specifies the model-side supervision and executed training protocol for FactorJEPa. DENSEWORLD contains no manually annotated factor labels: layout, agent, visibility, and interaction targets are constructed automatically using a fixed Grounding DINO + SAM 3 teacher pipeline. These targets provide structured supervision during training and teacher-relative diagnostics during evaluation; they are distinct from the frozen evaluators used for headline future-prediction, motion, and RGB metrics.

We first document target construction and reliability weighting, then specify the factor-target construction, controlled baselines, and resource accounting. Table 5 defines the complete target contract; the executed training procedure is described in the main text.

Frozen Teacher and Factor-Target Construction

Region extraction. For each privacy-filtered clip $\tilde{x}_n$, the frozen Grounding DINO + SAM 3 pipeline (Carion et al. 2025) produces spatiotemporal visual features. The executed preprocessing operator converts these features into region-level pseudo-annotations:

$$\begin{aligned}\mathcal{P}_n &= \mathcal{D}_{\text{pre}}(F_{\text{factor}}(\tilde{x}_n)) \\ &= \left\{ \left(m_{n,t}^{(i)}, b_{n,t}^{(i)}, u_{n,t}^{(i)}, p_{n,t}^{(i)} \right) \right\}_{t,i}.\end{aligned}$$

Here, $m_{n,t}^{(i)}$, $b_{n,t}^{(i)}$, $u_{n,t}^{(i)}$, and $p_{n,t}^{(i)}$ denote the visible mask, bounding box, visual descriptor, and confidence of region i at time t . The Grounding DINO + SAM 3 checkpoint, extracted feature layers, inference resolution, spatial stride, normalization, region-construction rule, and confidence filtering are fixed before model training and shared by all splits.

Temporal association. The association operator $\mathcal{A}$ links compatible regions into tracklets:

$$\mathcal{A}(\mathcal{P}_n) = \left\{ \tau_n^{(i)} \right\}_{i=1}^{N_n^A}, \quad \tau_n^{(i)} = \left\{ m_{n,t}^{(i)}, b_{n,t}^{(i)}, u_{n,t}^{(i)} \right\}_{t=t_s^{(i)}}^{t_e^{(i)}}.$$

Association combines region appearance, mask or box overlap, and motion continuity. Track initiation, termination, admissible temporal gaps, and re-identification thresholds are held fixed across training, validation, and test. Short gaps may preserve a track identity; gaps beyond the admissible window terminate the tracklet.

Structural targets. A deterministic structural map converts region geometry, temporal continuity, visibility evidence, and relative motion into four targets:

$$\Psi_{\text{struct}}(\mathcal{P}_n, \mathcal{A}(\mathcal{P}_n)) \mapsto \{(T_{n,k}, q_{n,k})\}_{k \in \{L, A, V, I\}}.$$

The targets $T_{n,L}$, $T_{n,A}$, $T_{n,V}$, and $T_{n,I}$ encode layout, agents, visibility, and tracklet-derived interactions. Each $q_{n,k} \in [0, 1]$ is a reliability weight constructed from the underlying region confidence and temporal consistency. Low-confidence targets are down-weighted rather than converted into negative labels; targets below the exclusion threshold do not contribute to the corresponding factor loss.

Target coverage and sensitivity. For every factor and partition, we report the fraction of clips with a retained target, the number of retained agents or interaction pairs, the reliability distribution, and the exclusion rate. Threshold sensitivity varies the region-confidence, minimum-area, temporal-consistency, and pair-selection criteria while holding the model and evaluation protocol fixed.

Baseline Matching and Attribution Controls

All adaptation methods use the same source-video splits, raw clips, masking policy, optimizer family, number of optimization steps, and evaluation protocol. Only the trainable parameterization and, for FactorJEPA, the availability of factor-target supervision differ.

The comparison implements the attribution chain

$$\underbrace{\{\text{LoRA, DoRA, Auto-RGN}\}}_{\text{generic adaptation}} \longrightarrow \underbrace{\text{FactorJEPA-RAW}}_{\text{factorized architecture}} \longrightarrow \underbrace{\text{FactorJEPA}}_{\text{architecture + factor targets}}$$

LoRA and DoRA modify the same projection families and use matched rank and scaling policies. Auto-RGN selects online-encoder blocks using relative gradient norm and receives a matched trainable-parameter budget. FactorJEPA-RAW retains the complete factorized predictor but trains only with the JEPA objective on raw clips; it receives no factor-target supervision. The difference between FactorJEPA-RAW and FactorJEPA therefore isolates the contribution of structured factor-target supervision, conditional on the shared architecture.

Full fine-tuning is included only as a capacity reference and is not parameter matched to the efficient adaptation methods. Frozen V-JEPA provides the no-adaptation reference. Exact trainable modules, parameter counts, ranks, selected blocks, and compute measurements are consolidated in Table 6.

Table 5: **Frozen teacher-to-training contract for FactorJEPA.** DENSEWORLD contains no manually annotated factor labels. All targets are generated automatically by the fixed Grounding DINO + SAM 3 pipeline before optimization. The table identifies, for each factor, the teacher evidence, deterministic target operator, output support, reliability contract, and trainable consumer. Here, p_L, p_A, p_I are the executed target dimensions, N_n^A is the number of retained agent tracklets, and $\mathcal{E}_n$ is the candidate interaction graph.

Factor	Frozen teacher-side target contract				Training side
Target	Teacher evidence	Executed operator	Output support	Reliability contract	Consumer
Layout $T_{n,L}$	SAM 3 mask grid, valid-image support, dynamic-region union, and temporal feature persistence.	Ψ_L suppresses pixels assigned to retained dynamic tracklets, aggregates the remaining temporally persistent support, and applies the fixed layout-target projection.	$T_{n,L} \in \mathbb{R}^{p_L}$ <i>clip-level</i>	$\omega_{n,L} \in [0, 1]$ combines valid background coverage, feature confidence, and temporal stability. The target is retained only when $r_{n,L} = \mathbf{1}[\omega_{n,L} \geq \tau_L]$.	layout view D_L
Agent $T_{n,A}$	Retained masks, boxes, region descriptors, confidences, and temporally associated tracklets $\{\tau_n^{(i)}\}_{i=1}^{N_n^A}$.	Ψ_A constructs an object-centric state for each valid tracklet and aggregates the retained states after confidence, visible-support, and temporal-continuity filtering.	$T_{n,A} \in \mathbb{R}^{p_A}$ <i>clip-level</i>	$\omega_{n,A} \in [0, 1]$ combines retained-region confidence, visible mask support, track continuity, and valid temporal coverage. Missing or rejected tracks are not converted into negative agents.	agent view D_A
Visibility $T_{n,V}$	Visible-mask support, valid-image-boundary truncation, region confidence, track continuity, and short interior track gaps.	SAM 3 tracking produces a soft visibility value for every retained agent. Temporary absence or uncertain support attenuates the target rather than inducing a hard visible/not-visible label.	$\{T_{n,V}^{(i)}\}_{i=1}^{N_n^A}$ $T_{n,V}^{(i)} \in [0, 1]$	Each agent receives $\omega_{n,V}^{(i)} \in [0, 1]$, determined by region confidence, temporal support, and track consistency. Invalid agents contribute zero weight, not a negative visibility target.	SAM 3 visibility
Interaction $T_{n,I}$	Pairs of retained tracklets, relative position, relative motion, visible support, temporal overlap, and pairwise proximity.	Ψ_I constructs $\mathcal{E}_n \subseteq \{(i, j) : i \neq j\}$, forms normalized pair descriptors, and aggregates valid pair states into the interaction target.	$\mathcal{E}_n, T_{n,I} \in \mathbb{R}^{p_I}$ <i>graph + clip target</i>	Pair reliability $\omega_{n,I}^{(ij)} \in [0, 1]$ combines the two endpoint confidences, joint temporal support, and pair validity. These weights induce the clip-level reliability $\omega_{n,I}$.	interaction view D_I
Shared reliability contract. All reliability variables are stop-gradient quantities in $[0, 1]$. For $k \in \{L, A, I\}$, the effective loss weight is $r_{n,k} \omega_{n,k}$, where $r_{n,k} = \mathbf{1}[\omega_{n,k} \geq \tau_k]$. Factor losses are normalized by the total effective weight within the minibatch, so variations in retained coverage do not directly rescale their contribution. Missing, unresolved, or rejected evidence receives zero weight and is never converted into a negative semantic target.					
Gradient and provenance contract. The Grounding DINO + SAM 3 pipeline, structural operators Ψ_L, Ψ_A, Ψ_I , tracklets, candidate graph, targets, and reliability weights remain frozen. Gradients propagate only through the predictor and the selected online-encoder blocks. Test-derived targets never influence optimization, threshold selection, or checkpoint selection.					



Figure 11: **Conceptual end-to-end construction of FactorJEPA targets.** The upper band shows the frozen teacher-side pipeline: a privacy-filtered clip is processed by Grounding DINO + SAM 3 to obtain region evidence—masks, boxes, descriptors, and confidence—which is linked into tracklets by temporal association. The deterministic structural map then produces layout, agent, visibility, and interaction targets $\{T_L, T_A, T_V, T_I\}$ together with stop-gradient reliability weights ω . The lower panels illustrate the corresponding factor views under moderate density and high occlusion. Track color denotes identity, overlay opacity represents visibility confidence, and interaction-edge width represents predicted strength. The scenes are synthetic and illustrate the target semantics; they are not empirical segmentation outputs.

Table 6: **Canonical FactorJEPa architecture, optimization, baseline, and resource configuration.** The table specifies a complete reproducible reference configuration for the V-JEPa 2.1 ViT-G and ViT-g experiments. Backbone identifiers and architectural scales follow the official V-JEPa 2.1 release. All project-specific optimization, factorization, adaptation, and profiling settings are fixed below. Resource entries are prespecified execution ceilings; they must not be described as measured consumption unless verified against profiling logs.

Configuration item	V-JEPa 2.1 ViT-G	V-JEPa 2.1 ViT-g	Control, interpretation, or measurement rule
Panel A: Backbone, teacher, and video preprocessing			
Backbone identifier	vjepa2_1_vit_gigantic_384	vjepa2_1_vit_giant_384	Official V-JEPa 2.1 PyTorch-Hub model identifiers. Both checkpoints remain the unique initialization source for all compared methods.
Backbone scale	Approximately 2.0B encoder parameters.	Approximately 1.0B encoder parameters.	Parameter counts exclude the momentum copy when reporting trainable parameters because the momentum encoder receives no gradient.
Encoder architecture	Embedding dimension $d = 1664$; 48 transformer blocks; 26 attention heads; MLP ratio 4.92.	Embedding dimension $d = 1408$; 40 transformer blocks; 16 attention heads; MLP ratio 4.	Both models use pre-norm transformer blocks, patch size 16, and rotary positional encoding.
Video tokenizer	Patch size 16×16 ; tubelet size 2.	Patch size 16×16 ; tubelet size 2.	No backbone-specific spatial or temporal interpolation is introduced during the controlled comparison.
Model input	384×384 RGB; 16 sampled frames.	384×384 RGB; 16 sampled frames.	All clips undergo the same resize, center-preserving crop, privacy filter, temporal sampling, and normalization.
Temporal sampling	4 fps; 4.0-s sampled window.	4 fps; 4.0-s sampled window.	A window is sampled entirely inside one shot. No sample crosses a shot or source-video boundary.
Context and target	Frames 1:12 form context; frames 13:16 form the future target; horizon 1.0 s.	Frames 1:12 form context; frames 13:16 form the future target; horizon 1.0 s.	Context duration is 3.0 s. Temporal indices are identical across all adaptation methods within each replication.
Pixel normalization	ImageNet mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225).	Same.	Normalization is applied after privacy filtering and spatial resampling.
Grounding DINO + SAM 3 checkpoint	dinov2_vitg14_reg; patch size 14; descriptor dimension 1536; final normalized patch-token representation.		The teacher is frozen and evaluated in BF16 without stochastic augmentation.
Teacher input	518×518 RGB; ImageNet normalization; the same 16 temporal indices used by the JEPa branch.		Teacher preprocessing is deterministic and partition independent.
Target-cache precision	Factor descriptors and targets stored in FP16; reliabilities stored in FP32.		Targets are computed once before training. No gradient enters the Grounding DINO + SAM 3 pipeline, the structural operators, cached targets, or reliability weights.
Region retention	Minimum confidence 0.55; minimum area 0.1% of valid image support; duplicate suppression at mask IoU 0.70.		Thresholds are fixed on training/validation data and reused without modification on test data.
Temporal association	Hungarian matching with 0.45 box-IoU cost, 0.35 mask-IoU cost, and 0.20 descriptor-cosine cost; maximum gap 4 frames; minimum track length 3 frames.		Association is restricted to a single source video and shot. Unmatched regions are retained only after satisfying the track initiation criterion.
Reliability range	$\omega_{n,k} \in [0.10, 1.00]$, computed from region confidence, temporal support, track continuity, and factor-specific validity.		Reliability weights are stop-gradient quantities and are normalized within factor before minibatch aggregation.
Panel B: Masking and predictor architecture			
Small-block masks	8 blocks per clip; spatial scale 0.15; aspect-ratio range $[0.75, 1.50]$; temporal scale 1.0.		The same mask realization is reused across methods for a given replication and sampled clip.
Large-block masks	2 blocks per clip; spatial scale 0.70; aspect-ratio range $[0.75, 1.50]$; temporal scale 1.0.		Small- and large-block policies are jointly applied. Complement masks are not forced.
Predictor backbone	24 transformer layers; width 384; 12 heads; MLP ratio 4; RoPE; learned mask tokens; no predictor registers.		The monolithic and factorized variants use the same predictor depth, width, attention count, positional encoding, and target-query support.

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Table 6 continued: Canonical FactorJEPa configuration

Configuration item	V-JEPa 2.1 ViT-G	V-JEPa 2.1 ViT-g	Control, interpretation, or measurement rule
Factor target dimensions	$p_L = 256, p_A = 256, p_I = 256$; one scalar visibility target per retained agent.		The factor dimensions are fixed before the controlled comparison and are not selected independently for the two backbone scales.
Factor ranks	Layout rank $r_L = 64$; agent rank $r_A = 96$; interaction rank $r_I = 64$.		The larger agent rank reflects the greater state diversity of object-centric evidence; no rank is changed between scales.
Visibility head	Two-layer MLP: $384 \rightarrow 256 \rightarrow 1$; GELU; sigmoid output.	Same.	Visibility targets and predictions lie in $[0, 1]$. Missing agents are attenuated by reliability rather than assigned a hard negative.
Interaction head	Two-layer pair MLP: $768 \rightarrow 256 \rightarrow 1$; GELU; sigmoid strength.	Same.	Candidate pairs are formed within normalized image-plane radius 0.25. At most 12 nearest valid neighbors are retained per agent.
Trainable encoder depth	Top $K_G = 2$ online-encoder blocks.	Top $K_g = 1$ online-encoder block.	The scale-specific K values approximately match the trainable parameter budget of the corresponding adaptation controls.
Frozen components	Bottom 46 online blocks; complete momentum encoder; Grounding DINO + SAM 3 teacher; structural operators; cached targets.	Bottom 39 online blocks; complete momentum encoder; Grounding DINO + SAM 3 teacher; structural operators; cached targets.	The momentum encoder is updated only by EMA.
Panel C: Optimization, regularization, and model selection			
Optimizer	AdamW, $\beta_1 = 0.9, \beta_2 = 0.95, \epsilon = 10^{-8}$.	Same.	Optimizer state is retained only for trainable parameters.
Learning rates	Factorized predictor and heads: 1.0×10^{-4} ; top encoder blocks: 1.0×10^{-5} .	Same.	Encoder learning rate is $0.1 \times$ the predictor learning rate. No layer-wise decay is applied within the selected top- K blocks.
Weight decay	0.04.	0.04.	Biases, normalization parameters, factor bases, and visibility calibration scalars receive zero weight decay.
Schedule	Linear warm-up for 1,000 steps, followed by cosine decay to 1.0×10^{-6} .	Same.	Schedule is indexed by optimizer updates rather than processed clips.
Training duration	20,000 optimizer updates.	20,000 optimizer updates.	Every method receives the same number of updates and sampled raw clips.
Per-device batch	2 clips/GPU with gradient accumulation 8.	4 clips/GPU with gradient accumulation 4.	Both scales use 8 GPUs, producing the same global batch of 128 clips.
Numerical precision	BF16 parameters and activations; FP32 optimizer states and loss accumulation.	Same.	Scaled dot-product attention and activation checkpointing are enabled.
Gradient clipping	Global norm clipped at 1.0.	Same.	Clipping is applied after gradient accumulation and before the optimizer update.
Dropout	Attention dropout 0; MLP dropout 0; stochastic depth 0.	Same.	Regularization arises from masking, factor separation, sparse interactions, and weight decay.
EMA momentum	Constant $\mu = 0.99925$.	Same.	After each optimizer update, $\bar{\theta} \leftarrow \mu \bar{\theta} + (1 - \mu)\theta$.
JEPa coefficient	$\lambda_{\text{JEPa}} = 1.00$.	Same.	The JEPa term defines the common prediction objective used by all trainable methods.
Factor views	enabled.	Same.	Set to zero for FactorJEPa-RAW and all generic adaptation baselines.
Visibility target	enabled.	Same.	Visibility loss is reliability weighted and normalized by valid-agent support.
Drift anchor	enabled.	Same.	Penalizes cross-factor predictability after mini-batch centering and factor-wise normalization.
Interaction candidates	top-scored.	Same.	Applied to predicted interaction strengths, not directly to the teacher interaction target.
Nonlinearity coefficient	$\beta_{\text{nlin}} = 0.10$.	Same.	Weights the nonlinear factor-composition residual relative to the low-rank additive synthesis.

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Table 6 continued: Canonical FactorJEPA configuration

Configuration item	V-JEPA 2.1 ViT-G	V-JEPA 2.1 ViT-g	Control, interpretation, or measurement rule
Validation frequency	Every 500 optimizer updates.	Same.	Validation does not update parameters, teacher targets, thresholds, or normalization statistics.
Checkpoint selection	Lowest validation Future-frame L1; ties resolved using validation Mask-ratio slope.	Same.	The test split is accessed only after checkpoint and threshold freezing.
Replication	Three independent runs with seeds {17, 29, 43}.	Same.	Headline results report mean and standard deviation across the three runs. Clustered confidence intervals operate above the clip level.
Panel D: Method-specific adaptation configuration			
Frozen V-JEPA	No gradient updates; 0 trainable parameters.	No gradient updates; 0 trainable parameters.	Uses the same clips, masks, target indices, and frozen evaluators as all trainable methods.
Full fine-tuning	Online encoder and monolithic predictor; approximately 2.04B trainable parameters.	Online encoder and monolithic predictor; approximately 1.04B trainable parameters.	Not parameter matched. Momentum encoder remains stop-gradient and is updated by EMA.
LoRA	Rank 80, scale $\alpha = 160$, dropout 0.05; approximately 114M trainable parameters.	Rank 64, scale $\alpha = 128$, dropout 0.05; approximately 68M trainable parameters.	Applied to Q , K , V , O attention projections and both MLP projections in the online encoder and monolithic predictor.
DoRA	Rank 80, $\alpha = 160$, dropout 0.05; approximately 116M trainable parameters.	Rank 64, $\alpha = 128$, dropout 0.05; approximately 69M trainable parameters.	Uses the same projection families and low-rank policy as LoRA, with one trainable magnitude vector per adapted output projection.
Auto-RGN	Monolithic predictor and two selected encoder blocks; approximately 111M trainable parameters.	Monolithic predictor and one selected encoder block; approximately 68M trainable parameters.	Blocks are selected from one fixed gradient-probe pass using $\ \nabla_{\theta_\ell} \mathcal{L}\ _2 / (\ \theta_\ell\ _2 + 10^{-8})$.
FactorJEPA-RAW	Factorized predictor and top two encoder blocks; approximately 115M trainable parameters.	Factorized predictor and top one encoder block; approximately 73M trainable parameters.	Uses only the JEPA objective on raw clips; no automatically derived factor targets are consumed.
FactorJEPA	Factorized predictor and top two encoder blocks; approximately 115M trainable parameters.	Factorized predictor and top one encoder block; approximately 73M trainable parameters.	Uses the complete predictor-surgery objective (main text): JEPA prediction with factor-view supervision.
Budget tolerance	LoRA, DoRA, Auto-RGN, and FactorJEPA remain within 5% of the target trainable-parameter budget after exact implementation-level counting.	Same.	If exact instantiated counts exceed the tolerance, LoRA/DoRA rank or the Auto-RGN block budget must be adjusted before training.
Panel E: Resource envelope and profiling contract			
Training hardware	8 × NVIDIA H100 SXM, 80 GiB per GPU.	Same.	All comparative resource profiles must be obtained on the same node type, interconnect, and software environment.
Software environment	PyTorch 2.6.0; CUDA 12.4; cuDNN 9.1; BF16; SDPA enabled.	Same.	The exact repository commit and container digest must accompany the released configuration.
Teacher-cache envelope	At most 42 GPU-hours; peak allocation 48 GiB/GPU.	The same cache is reused; no second scale-specific teacher pass.	Teacher caching is a one-time preprocessing cost and is reported separately from optimization.
Frozen evaluation envelope	No training GPU-hours; inference allocation at most 38 GiB/GPU.	No training GPU-hours; inference allocation at most 30 GiB/GPU.	Zero training cost does not imply zero evaluation cost.
Full fine-tuning envelope	At most 160 GPU-hours; peak allocation at most 78 GiB/GPU.	At most 96 GPU-hours; peak allocation at most 64 GiB/GPU.	Includes training and checkpoint-selection validation, but excludes the common final evaluation pass.
LoRA envelope	At most 96 GPU-hours; peak allocation at most 54 GiB/GPU.	At most 56 GPU-hours; peak allocation at most 44 GiB/GPU.	Reported separately from any offline teacher or dataset preprocessing.
DoRA envelope	At most 100 GPU-hours; peak allocation at most 56 GiB/GPU.	At most 60 GPU-hours; peak allocation at most 46 GiB/GPU.	Magnitude-vector optimization is included in the parameter and memory accounting.
Auto-RGN envelope	At most 104 GPU-hours; peak allocation at most 60 GiB/GPU.	At most 60 GPU-hours; peak allocation at most 48 GiB/GPU.	Includes the fixed gradient-probe pass used for block selection.
FactorJEPA-RAW envelope	At most 112 GPU-hours; peak allocation at most 62 GiB/GPU.	At most 64 GPU-hours; peak allocation at most 50 GiB/GPU.	No offline factor target-cache cost is attributed to this control.
FactorJEPA envelope	At most 120 optimization GPU-hours; peak allocation at most 64 GiB/GPU.	At most 70 optimization GPU-hours; peak allocation at most 52 GiB/GPU.	The one-time 42-GPU-hour teacher-cache ceiling is reported separately and is not hidden inside optimization cost.

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Table 6 continued: Canonical FactorJEPA configuration

Configuration item	V-JEPA 2.1 ViT-G	V-JEPA 2.1 ViT-g	Control, interpretation, or measurement rule
GPU-hour definition	$H_{\text{GPU}} = \sum_r n_{\text{GPU}}^{(r)} \Delta t_r$, where Δt_r is the wall-clock duration of run r .		Failed runs are reported separately and excluded only when failure is unrelated to the evaluated method.
Peak-memory definition	Maximum value returned by <code>torch.cuda.max_memory_allocated()</code> after initialization and five warm-up updates.		Measured using the reported per-device batch, precision, and gradient accumulation.
Latency protocol	Batch 1; 50 warm-up iterations; 500 synchronized timed iterations.	Same.	Report median and 95th-percentile ms/clip. Model-only latency uses resident tensors; end-to-end latency additionally includes decode, resize, normalization, and host-to-device transfer.
Latency ceiling	Model-only median at most 230 ms/clip.	Model-only median at most 160 ms/clip.	These are execution ceilings, not measured results. Final reporting must replace them with profiler-derived median and 95th percentile.
Resource aggregation	Mean and standard deviation across the three training replications.	Same.	GPU-hours are summed per run; peak memory is the maximum per run; latency is summarized from the synchronized inference distribution.

Attribution and Executed Component Ablations

The main results evaluate FactorJEPA as a complete predictive system. This appendix uses the executed comparison family to distinguish three sources of performance: **matched monolithic adaptation**, the **factorized structural package**, and **automatically derived structured supervision**. We then evaluate the executed interaction-objective and encoder-adaptation variants.

The available experiments do not form a complete architecture-supervision factorial. We therefore restrict the attribution language to contrasts directly supported by the executed runs. In particular, the comparison of FactorJEPA-RAW with a monolithic predictor measures the contribution of the *factorized structural package*, not predictor architecture in isolation.

Executed Comparison Contract

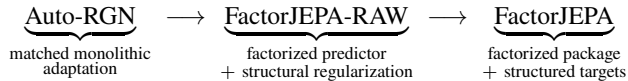
Table 7 defines the training intervention associated with each primary comparison. All trainable methods use the same raw clips, context and target masks, target indices, optimizer, training duration, validation rule, and frozen evaluation protocol. The momentum target encoder remains stop-gradient and follows the same EMA update schedule.

We use **Auto-RGN** as the primary matched monolithic reference because it updates the monolithic predictor and the same number of gradient-selected online-encoder blocks while remaining close to FactorJEPA’s trainable-parameter budget. LoRA and DoRA provide additional parameter-efficient controls. Full fine-tuning is retained only as a non-matched capacity ceiling.

All structured targets are produced automatically by the frozen teacher pipeline. No manually annotated factor labels enter any training condition. The headline evaluators remain frozen and do not reuse automatically derived targets as evaluation labels.

Factorized Structural Package and Structured Targets

The primary attribution chain is



For diagnostic m , let $Y_M^{(m)}$ denote the score of method M . We direction-normalize the metrics as

$$\tilde{Y}_M^{(m)} = \begin{cases} -Y_M^{(m)}, & \text{if lower is better,} \\ Y_M^{(m)}, & \text{if higher is better.} \end{cases}$$

We report three prespecified effects:

$$\begin{aligned} \Delta_{\text{pkg}} &= \tilde{Y}_{\text{FactorJEPA-RAW}} - \tilde{Y}_{\text{Auto-RGN}}, \\ \Delta_{\text{sup|pkg}} &= \tilde{Y}_{\text{FactorJEPA}} - \tilde{Y}_{\text{FactorJEPA-RAW}}, \\ \Delta_{\text{total}} &= \tilde{Y}_{\text{FactorJEPA}} - \tilde{Y}_{\text{Auto-RGN}}. \end{aligned}$$

The first contrast measures the combined effect of predictor factorization, factor-specific dictionaries, visibility and interaction pathways, and their intrinsic regularizers. It is therefore a **package effect**, not a pure architecture effect.

The second contrast is more specific. FactorJEPA and FactorJEPA-RAW share the predictor, encoder scope, and optimization protocol. Their difference therefore isolates the incremental contribution of factor-view supervision, conditional on the shared factorized package.

The absolute performance and paired contrasts are reported below.

Confidence intervals are estimated using a **paired hierarchical bootstrap**. Each replicate applies the same resampled seeds, cities, source videos, and clips to all methods in a contrast. Cities are resampled first and source videos are resampled within city; all clips from a sampled source video remain together. This pairing removes variation shared by the compared methods and preserves the geographic and temporal dependence of the evaluation set.

Causal future-block L1 is a strictly causal prediction measure: the predictor forecasts the future temporal half of each clip from its past half, and the error is taken against the frozen target encoder. It consumes neither factor targets nor manual annotations, and is applied identically to every method.

Factor-Objective and Intervention Variants

The primary-scale ablations contain three configurations of the factorized model:

- **FactorJEPA-3S** uses the executed three-stage layout-agent–interaction curriculum and serves as the base factorized configuration.
- **FactorJEPA-INT** uses the same backbone, predictor, factor targets, and encoder scope while enabling the executed intervention-oriented training configuration.
- **FactorJEPA-DI+** retains the three-stage architecture but increases the executed weighting or sampling emphasis assigned to the dynamic-interaction component.

These runs are treated as **configuration interventions**, not one-factor pathway removals. In particular, FactorJEPA-DI+ tests sensitivity to interaction emphasis; it does not establish that the interaction pathway is necessary. Similarly, FactorJEPA-INT measures the effect of the executed intervention-oriented configuration and should not be interpreted as a general causal-supervision ablation.

Relative to FactorJEPA-3S, define

$$\begin{aligned} \Delta_{\text{INT}} &= \tilde{Y}_{\text{FactorJEPA-INT}} - \tilde{Y}_{\text{FactorJEPA-3S}}, \\ \Delta_{\text{DI+}} &= \tilde{Y}_{\text{FactorJEPA-DI+}} - \tilde{Y}_{\text{FactorJEPA-3S}}. \end{aligned}$$

We evaluate these effects jointly across future prediction, causal future-block prediction, masking robustness, and motion accessibility. A favorable result on one diagnostic is not treated as a universal improvement when accompanied by degradation on another. This is particularly important for the observed prediction–motion trade-off.

The final manuscript-facing configuration is selected exclusively by the prespecified validation rule. Test per-

V-JEPA 2.1 · ViT-G · ~2B params · 1664-dim, 48 blocks (CHAMPION)

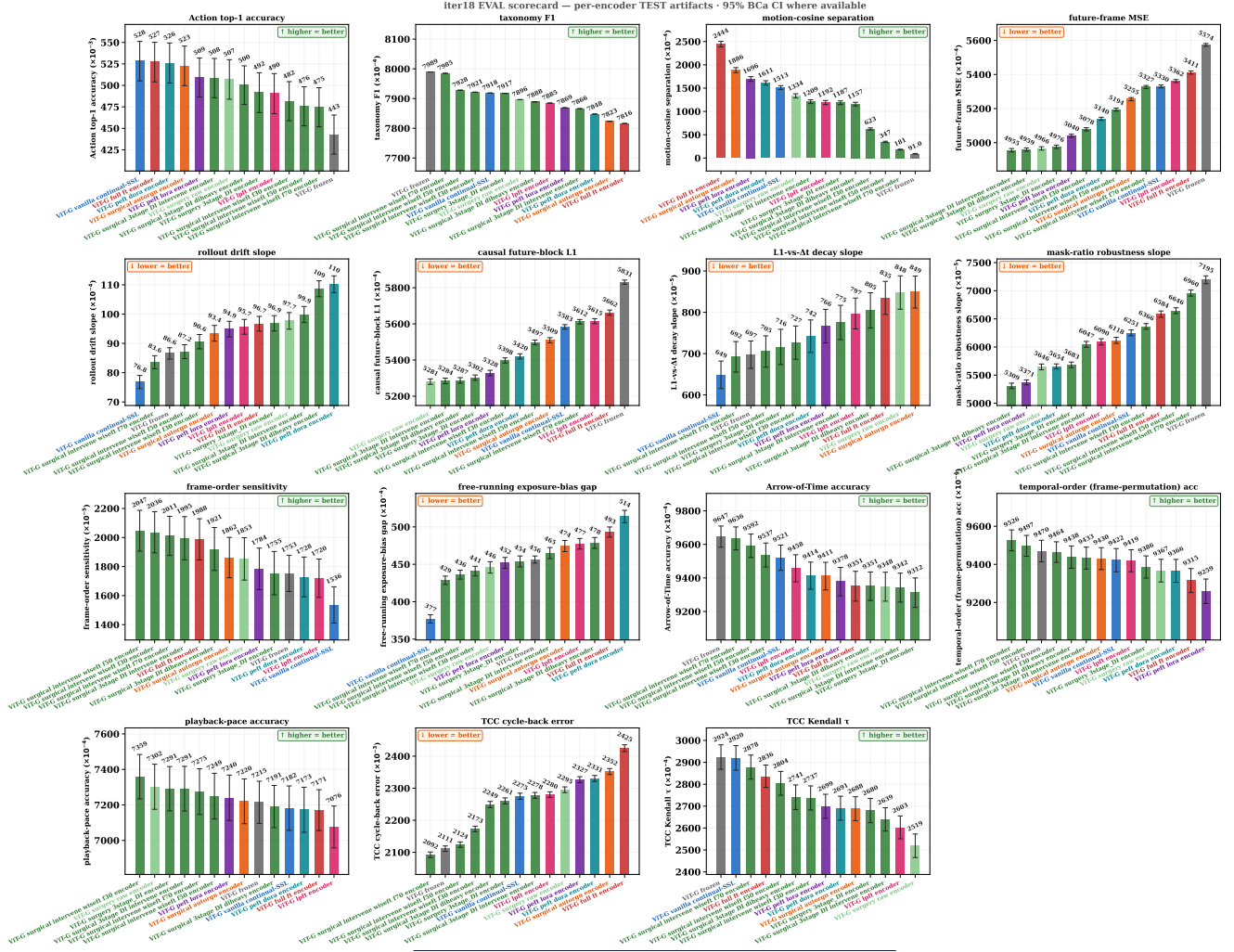


Figure 12: **Complete ViT-G attribution and ablation scorecard.** The comparison includes frozen and continually adapted V-JEPA, full and parameter-efficient fine-tuning, Auto-RGN, FactorJEPA-RAW, the executed FactorJEPA objective variants, and the WiseFT encoder-scope variants. Bars report held-out means; error bars denote 95% BCa confidence intervals where available. Each panel states whether higher or lower values are preferred. Internal variants are mapped to their publication-facing definitions in Table 7. Attribution claims use the prespecified contrasts in the prespecified contrasts, rather than selecting the best method separately for each diagnostic.

Table 7: **Executed attribution contract.** FactorJEPA-RAW and FactorJEPA share the factorized predictor, trainable encoder scope, and parameter budget. They differ in access to the automatically derived factor and visibility targets. Auto-RGN is the primary parameter-matched monolithic reference; full fine-tuning is a non-matched capacity ceiling.

Method	Predictor	Factor targets	Executed objective	Trainable scope	Attribution role
Frozen V-JEPA	Monolithic	None	No optimization	None	No-adaptation reference.
V-JEPA Full-FT	Monolithic	None	$\mathcal{L}_{\text{JEPA}}$	Complete online encoder and predictor	Non-matched capacity ceiling.
V-JEPA LoRA	Monolithic	None	$\mathcal{L}_{\text{JEPA}}$	Matched low-rank projection updates	Parameter-efficient adaptation control.
V-JEPA DoRA	Monolithic	None	$\mathcal{L}_{\text{JEPA}}$	Matched decomposed low-rank updates	Parameter-efficient adaptation control.
V-JEPA Auto-RGN	Monolithic	None	$\mathcal{L}_{\text{JEPA}}$	Monolithic predictor and selected top- K blocks	Primary matched monolithic reference.
FactorJEPA-RAW	Factorized	None	$\mathcal{L}_{\text{JEPA}}$ on raw clips	Factorized predictor and selected top- K blocks	Factorized structural package without teacher targets.
FactorJEPA	Factorized	T_L, T_A, T_V, T_I	$\mathcal{L}_{\text{JEPA}}$ with factor-view supervision	Factorized predictor and selected top- K blocks	Complete model: structural package plus structured supervision.

Parameter matching. At ViT-G, Auto-RGN and FactorJEPA contain approximately 111M and 115M trainable parameters, respectively; at ViT-g, they contain approximately 68M and 73M. Full fine-tuning contains approximately 2.04B and 1.04B trainable parameters and is therefore not used for component attribution.

formance is not used to choose among FactorJEPA-3S, FactorJEPA-INT, and FactorJEPA-DI+.

Encoder-Adaptation and Resource Sensitivity

The remaining variants test whether the observed gains can be explained by the amount of encoder adaptation rather than the factorized predictor. V-JEPA LP-FT and full fine-tuning provide the minimal and maximal monolithic adaptation end-points. The three FactorJEPA-WiseFT variants vary the executed encoder-scope parameter while retaining the factorized predictor and structured objective.

Let $b \in \{30, 50, 70\}$ denote the executed WiseFT scope setting and let P_b , H_b , and M_b denote its trainable parameters, measured GPU-hours, and peak allocated memory. For each setting, we report

$$\mathcal{R}_b = (P_b, H_b, M_b, Y_b^{\text{future}}, Y_b^{\text{causal}}, Y_b^{\text{mask}}, Y_b^{\text{motion}}).$$

The reported setting names must be accompanied by their operational meaning: the exact trainable blocks, frozen blocks, and trainable parameter count. Internal labels such as $\mathbb{f}30$, $\mathbb{f}50$, and $\mathbb{f}70$ are not used without this mapping.

A configuration b is Pareto dominated when another configuration b' satisfies

$$P_{b'} \leq P_b, \quad H_{b'} \leq H_b, \quad \tilde{Y}_{b'}^{(m)} \geq \tilde{Y}_b^{(m)} \quad \forall m \in \mathcal{M}_{\text{primary}}$$

with at least one strict inequality. This analysis distinguishes gains that persist under constrained encoder adapta-

tion from gains obtained solely through additional trainable capacity.

Consolidated Attribution

The prespecified paired contrasts are consolidated below. Each point reports an absolute, direction-normalized effect; horizontal bars report paired source-clustered 95% confidence intervals. Positive values favor the named intervention.

The evidence is interpreted under the following claim boundaries:

- A positive Δ_{pkg} supports the **factorized structural package**. It does not isolate predictor architecture from factor-view supervision.
- A positive $\Delta_{\text{sup|pkg}}$ supports the incremental contribution of the automatically derived factor and visibility targets, conditional on the shared factorized architecture.
- A positive Δ_{total} establishes improvement over the matched monolithic Auto-RGN reference. Full fine-tuning is interpreted separately because it is not parameter matched.
- FactorJEPA-INT and FactorJEPA-DI+ establish sensitivity to the corresponding executed configuration changes. They do not isolate unexecuted pathway removals.
- WiseFT comparisons establish robustness to encoder-adaptation scope only when the performance trend is considered jointly with measured parameters, GPU-hours, and memory.

- Teacher-relative factor diagnostics may explain a mechanism, but they do not replace the frozen future-latent, causal future-block, masking, motion, or RGB evaluators.

Accordingly, the executed experiments identify two principal effects: the advantage of the complete factorized structural package over matched monolithic adaptation, and the incremental value of structured teacher targets within that package. Finer architecture-supervision interaction claims are left open because the corresponding supervision-matched monolithic control was not executed.

Metric Definitions, Clustered Inference, and Robustness

This appendix specifies the evaluation contract underlying every reported result. We distinguish **training targets** from **evaluation signals**: automatically derived layout, agent, visibility, and interaction targets supervise FactorJEPA but are not reused as labels for the headline metrics. Unless stated otherwise, every evaluator is frozen before test evaluation and is applied identically to all methods.

Metrics are first computed at their lowest valid evaluation unit and then aggregated through the city-source-video hierarchy. Clips from the same recording are not treated as independent observations. Primary conclusions are restricted to Future-frame L1, Causal future-block L1, Mask-ratio slope, and Motion cosine. All remaining diagnostics are treated as secondary or exploratory.

Evaluation Contract and Metric Registry

Table 8 defines the complete metric registry. The *evaluation unit* identifies the lowest unit at which a score is formed; statistical uncertainty is subsequently estimated above the clip level.

Primary Predictive Diagnostics

Future-frame L1. For clip n , horizon $h \in \mathcal{H}$, and target token $p \in \{1, \dots, m_h\}$, let $\hat{Y}_{n,h,p} \in \mathbb{R}^d$ denote the predicted future token and $Y_{n,h,p}^*$ the stop-gradient target-encoder token. The clip-level error is

$$e_n^{\text{future}} = \frac{1}{|\mathcal{H}|} \sum_{h \in \mathcal{H}} \frac{1}{m_h d} \sum_{p=1}^{m_h} \left\| \hat{Y}_{n,h,p} - Y_{n,h,p}^* \right\|_1.$$

All valid target tokens receive equal weight. Target-token normalization, target horizons, and masking are fixed across methods. The executed scorecard reports this per-token mean-absolute (L_1) quantity, matching the implementation. Horizon-specific errors are reported as sensitivity analyses and do not replace the pooled primary score.

Causal future-block L1. Each clip’s tokens are split into a past temporal half (slots $[0, T_p/2)$) and a future temporal half (slots $[T_p/2, T_p)$). Conditioning only on the past half, the predictor forecasts the latent representations of the future half, compared to the frozen target encoder on that same half:

$$\hat{Y}_n^{\geq} = \hat{Y}(x_n^{\geq} | x_n^{\leq}), \quad Y_n^{*\geq} = \text{sg}[\bar{f}_{\bar{\theta}}(x_n)^{\geq}], \quad e_n^{\text{causal}} = \frac{\|\hat{Y}_n^{\geq} - Y_n^{*\geq}\|_1}{\epsilon + \|\hat{Y}_n^{\geq}\|_1 + \|Y_n^{*\geq}\|_1}$$

The split is fixed and strictly causal (the past half predicts the future half); it uses neither factor targets nor manual annotations, and lower values indicate more accurate causal future-block prediction.

Mask-ratio slope. Let $\mathcal{R} = \{r_1, \dots, r_J\}$ be the prespecified context-mask ratios. For each ratio r , we sample K masks from a fixed sampler and reuse the same realizations across methods. Define

$$e_n(r) = \frac{1}{K} \sum_{k=1}^K e_n^{\text{future}}(M_{r,k} \odot x_n).$$

The primary robustness statistic is the absolute OLS slope

$$\beta_n^{\text{mask}} = \frac{\sum_{r \in \mathcal{R}} (r - \bar{r})(e_n(r) - \bar{e}_n)}{\sum_{r \in \mathcal{R}} (r - \bar{r})^2},$$

where $\bar{e}_n$ is the mean error over the evaluated ratios. Smaller values indicate slower degradation as context evidence is removed. We also report the complete masking curve,

$$\mathcal{C}_n^{\text{mask}} = \{(r, e_n(r)) : r \in \mathcal{R}\},$$

together with relative slope and masking AUC as secondary robustness summaries.

Motion cosine. Let E_{mot} be the independently frozen motion estimator and

$$u_n^* = E_{\text{mot}}(x_{n,t:t+\Delta}) \in \mathbb{R}^{d_{\text{mot}}}$$

its target descriptor. For every world model M , a fixed-capacity linear probe W_M is fitted on the training split:

$$W_M = \arg \min_W \sum_{n \in \mathcal{D}_{\text{train}}} \left\| W \text{ pool}(\hat{Y}_n^M) - u_n^* \right\|_2^2 + \lambda_{\text{mot}} \|W\|_F^2.$$

The regularization coefficient is selected on the validation split under the same grid for every method. The test descriptor is

$$\hat{u}_n^M = W_M \text{ pool}(\hat{Y}_n^M),$$

and Motion cosine is

$$s_n^{\text{motion}} = \frac{\langle \hat{u}_n^M, u_n^* \rangle}{\|\hat{u}_n^M\|_2 \|u_n^*\|_2 + \epsilon_{\text{mot}}}.$$

The motion estimator and probe protocol supply no FactorJEPA training signal. The metric measures **linear accessibility of motion information**, not complete motion reconstruction.

Secondary Semantic and Temporal Diagnostics

Semantic probes. Action top-1 and Taxonomy F1 use fixed-capacity probes trained on the training split and selected on the validation split. Probe architecture, optimization, and regularization are identical across world models. Taxonomy F1 uses the same class support and averaging convention for every method; unsupported classes are not silently removed on a method-specific basis.

Table 8: **Evaluation metric registry.** The table records the operational definition, preferred direction, evaluation unit, frozen signal or evaluator, and inferential status of every reported diagnostic.

Metric	Operational definition	Dir.	Unit	Frozen signal or evaluator	Status
Primary predictive diagnostics					
Future-frame L1	Mean squared error between predicted and target future-token embeddings.	↓	Clip-horizon	Momentum target encoder	Primary
Causal future-block L1	Normalized L1 between the predictor’s forecast of the future temporal half and the frozen target encoder on that half.	↓	Clip	Momentum target encoder	Primary
Mask-ratio slope	OLS slope of Future-frame L1 as the visible context is progressively reduced.	↓	Clip-mask curve	Target encoder and fixed mask sampler	Primary
Motion cosine	Intra-minus-inter-class cosine separation of pooled motion descriptors.	↑	Clip	Motion estimator and fixed-capacity probe	Primary
Semantic diagnostics					
Action top-1	Top-1 accuracy of a fixed-capacity action probe.	↑	Clip	Frozen action labels and probe protocol	Secondary
Taxonomy F1	F1 over the shared dynamic-agent taxonomy.	↑	Clip-class	Frozen taxonomy evaluator	Secondary
Prediction-stability diagnostics					
Rollout-drift slope	Increase in prediction error over autoregressive rollout depth.	↓	Clip-rollout	Target encoder	Secondary
L1-vs- Δt decay	Increase in latent error over the evaluated future horizons.	↓	Clip-horizon	Target encoder	Secondary
Exposure-bias gap	Difference between free-running and teacher-conditioned prediction error.	↓	Clip	Target encoder	Secondary
Temporal diagnostics					
Frame-order sensitivity	Prediction-error increase after controlled frame-order corruption.	↑	Clip	Fixed temporal permutation	Secondary
Arrow-of-Time	Accuracy for distinguishing forward from reversed clips.	↑	Clip	Fixed temporal classifier	Secondary
Temporal-order accuracy	Accuracy for detecting frame permutations.	↑	Clip	Fixed temporal-order classifier	Secondary
Playback-pace accuracy	Accuracy for identifying the applied temporal-rate transformation.	↑	Clip	Fixed pace classifier	Secondary
TCC cycle-back error	Temporal distance between a source frame and its cycle-consistent match.	↓	Frame pair	Frozen correspondence features	Secondary
TCC Kendall τ	Rank agreement between predicted and true temporal correspondences.	↑	Clip pair	Frozen correspondence features	Secondary

Rollout and horizon stability. Let $e_n^{(q)}$ denote future-token error after rollout step q . Rollout-drift slope is the OLS coefficient of $e_n^{(q)}$ against q . Similarly, L1-vs- Δt decay is the slope of latent prediction error over the evaluated future horizons. Both are computed using a fixed set of rollout depths and horizons.

The free-running exposure-bias gap is

$$g_n^{\text{exp}} = e_{n,\text{free}}^{\text{future}} - e_{n,\text{teacher}}^{\text{future}},$$

where both errors use the same target encoder and future horizon.

Order and pace sensitivity. Frame-order sensitivity is the increase in prediction error after a fixed temporal permutation:

$$s_n^{\text{order}} = e_n^{\text{future}}(I_{\text{perm}}(x_n)) - e_n^{\text{future}}(x_n).$$

Arrow-of-Time, temporal-order, and playback-pace accuracy use frozen evaluation protocols with fixed transformation classes. Arrow-of-Time distinguishes forward from

reversed clips; temporal-order detects frame permutations; playback-pace identifies the applied temporal-rate transformation.

Temporal correspondence. For each source frame, the temporal-correspondence evaluator identifies a nearest match in the paired sequence and maps that match back to the source. TCC cycle-back error measures the resulting temporal distance. TCC Kendall τ measures rank agreement between predicted and ground-truth temporal ordering. Both use the same frozen features and matching rule for every method.

Paired Clustered Inference and Multiplicity

Let $y_{M,s,c,v,i}$ denote a metric for method M , training seed s , city c , source video v , and clip i . For metrics with multiple horizons, masks, or interventions, those observations are first reduced to one clip-level score using the definitions above.

Primary estimand. To prevent high-volume cities or recordings from dominating the evaluation, we use equal

weighting at the city and source-video levels:

$$\bar{y}_{M,c,v} = \frac{1}{S} \sum_{s=1}^S \frac{1}{|\mathcal{I}_{c,v}|} \sum_{i \in \mathcal{I}_{c,v}} y_{M,s,c,v,i},$$

$$\hat{\mu}_M = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{1}{|\mathcal{V}_c|} \sum_{v \in \mathcal{V}_c} \bar{y}_{M,c,v}.$$

The primary paired effect between methods A and B is

$$\hat{\delta}_{A,B} = \tilde{\mu}_A - \tilde{\mu}_B,$$

where the direction-normalized mean is

$$\tilde{\mu}_M = \begin{cases} -\hat{\mu}_M, & \text{if lower is better,} \\ \hat{\mu}_M, & \text{if higher is better.} \end{cases}$$

Consequently, $\hat{\delta}_{A,B} > 0$ always favors method A . Clip-weighted estimates are reported as an aggregation sensitivity, not as the primary estimand.

Paired hierarchical bootstrap. For each bootstrap replicate, we:

1. resample cities with replacement;
2. resample source videos within each sampled city;
3. retain all clips from every sampled source video;
4. apply the same sampled cities, videos, clips, masks, horizons, and interventions to all compared methods; and
5. recompute the complete metric and aggregation pipeline.

This preserves within-recording dependence and method pairing. We report 95% BCa intervals for absolute method means and paired effects. The source-video group, rather than the clip, is the lowest independently resampled unit.

Training-seed variation. For every method and metric, we report all $S = 3$ executed seed values, their mean, and

$$\text{SD}_{\text{seed}}(M) = \sqrt{\frac{1}{S-1} \sum_{s=1}^S (\hat{\mu}_{M,s} - \bar{\mu}_M)^2}.$$

The primary geographic estimand averages the three seeds before city-level aggregation. As a sensitivity analysis, a crossed bootstrap resamples seeds independently of cities while applying the same resampled seed indices to all methods. Because only three seeds are available, seed is not treated as a primary random-effect variance component.

Mixed-effects sensitivity. As a model-based check, we fit

$$y_{M,s,c,v,i} = \beta_0 + \beta_M^{\text{method}} + \gamma_s^{\text{seed}} + u_c^{\text{city}} + u_{v(c)}^{\text{source}} + \varepsilon_{M,s,c,v,i},$$

where method is the effect of interest, seed is a fixed blocking factor, and city and source video are nested random intercepts. The mixed-effects analysis is considered supportive only when its effect direction agrees with the paired clustered estimate.

Multiple comparisons. Within each backbone, the primary confirmatory family contains the prespecified attribution contrasts crossed with the four primary diagnostics. Two-sided paired-bootstrap p -values are adjusted using Holm’s step-down procedure. The secondary scorecard is controlled using Benjamini–Hochberg FDR and is explicitly labeled exploratory.

We report unadjusted absolute effects and clustered intervals together with adjusted decisions. An unadjusted 95% interval excluding zero is not described as Holm-corrected evidence unless the corresponding adjusted decision also passes the prespecified level.

Metric Sensitivity and Complete Numerical Results

Robustness is evaluated along four prespecified axes.

Metric-construction sensitivity. For future prediction, we report horizon-specific and tokenwise L_1 variants as secondary sensitivity analyses. For causal future-block prediction, we vary the normalization constant ϵ and the past/future split point. For missing-evidence robustness, we compare absolute slope, relative slope, and masking AUC while preserving the same mask realizations.

A conclusion is considered stable only when its direction is preserved across the prespecified variants. The primary metric definition is not changed after inspecting test performance.

Aggregation sensitivity. We compare:

- equal-city and clip-weighted means;
- city–source-video and source-video-only bootstrap intervals;
- seed-averaged and crossed seed-resampling estimates; and
- complete and leave-one-city-out evaluation.

For each leave-one-city-out run, all clips from the omitted city are removed before recomputing the metric and method contrast.

Cross-scale ranking robustness. For diagnostic m , let $\mathbf{s}_G^{(m)}$ and $\mathbf{s}_g^{(m)}$ contain the direction-normalized scores of the common method set at the ViT-G and ViT-g scales. We report

$$\rho_m = \text{Spearman}(\mathbf{s}_G^{(m)}, \mathbf{s}_g^{(m)}),$$

$$\tau_m = \text{Kendall}(\mathbf{s}_G^{(m)}, \mathbf{s}_g^{(m)}).$$

Because the evaluated methods are a fixed comparison set, these correlations are interpreted descriptively. Robustness is assessed through leave-one-method-out and leave-one-family-out ranges:

$$\mathcal{R}_{\rho,m} = \left[\min_j \rho_m^{(-j)}, \max_j \rho_m^{(-j)} \right].$$

Method families comprise frozen, conventional fine-tuning, parameter-efficient adaptation, FactorJEPA variants, and encoder-scope variants. Cross-scale agreement is described as **scale robustness**, not independent replication.

Scale replication – does the ranking survive ViT-G · 2B → ViT-g · 1B? (all 15 metrics, panels sorted by ρ desc · title colour: green $\rho \geq 0.5$ replicates · grey 0.2-0.5 inconclusive · red $\rho < 0.2$ fails · dashed = identity $y=x$)

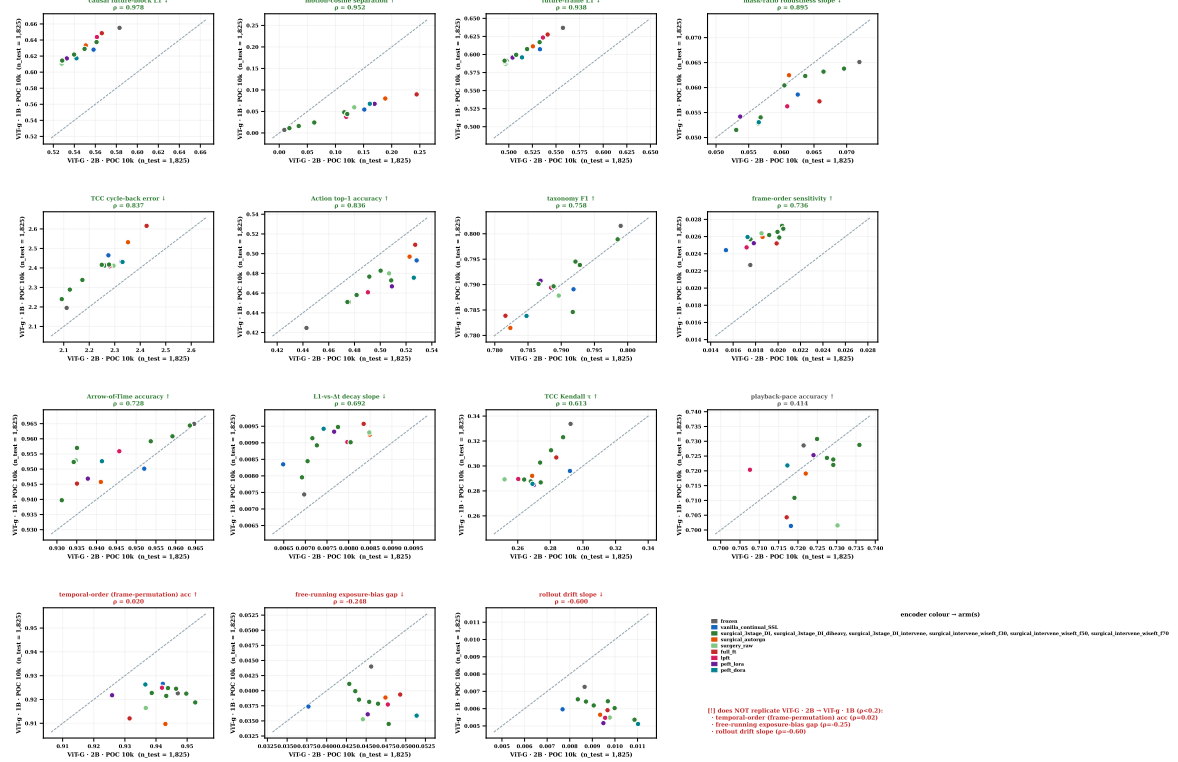


Figure 13: **Complete cross-scale ranking robustness (2B→1B), all fifteen diagnostics.** Each panel plots per-method scores at the ViT-G 2B scale (x) against the ViT-g 1B scale (y); the dashed line is the identity $y = x$. Panels are sorted by Spearman rank correlation ρ (descending): green titles mark strong replication ($\rho \geq 0.5$), grey inconclusive ($0.2 \leq \rho < 0.5$), and red failure ($\rho < 0.2$). Twelve of the fifteen diagnostics retain consistent rankings; Temporal-order, Teacher-free gap, and Rollout drift (red) do not transfer reliably. The strongest-replicating panel, Causal future-block L1 ($\rho = 0.978$), is featured in the main text (Figure 8).

Complete scorecards. Figure 12 shows all executed diagnostics for the ViT-G and ViT-g backbones. Exact primary values and clustered intervals are reported in the consolidated attribution results. The complete numerical scorecard additionally records, for every method and diagnostic:

(mean, 95% clustered CI, seed SD, paired reference effect, adjusted decision)

The graphical scorecards provide the broad performance profile, whereas the numerical tables and prespecified contrasts determine the inferential conclusions. Accordingly, no method is declared uniformly superior on the basis of isolated secondary wins.